

Risk Analysis Probability of Default: A Stochastic Simulation Model

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Comments Welcome

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Abstract

We present a stochastic simulation model for estimating forward-looking corporate probability of default and loss given default. We formulate the model in a discrete time frame, apply capital-budgeting techniques to define the relationships that identify the default condition, and solve the model by Monte Carlo simulation. First, we present the model; then we show how to extend the model to estimate company specific loss given default, expected loss and unexpected loss as well. Subsequently, we compare the RAPD model with option/contingent models, inasmuch as both models use the same definition of the event of default. The focus of this paper is on the theoretical and modeling aspects of the new methodological approach proposed; however we also present an application of the method that represents just one example of its possible implementations, and the results of a comparative test (covering RAPD, Altman Z-score, two option-contingent models and S&P ratings) which in our opinion constitutes a preliminary positive empirical support of the validity of the RAPD Approach. Conclusive remarks end the paper.

Key words: capital budgeting, corporate valuation, credit risk, discounted cash flow (DCF), expected loss, loss given default (LGD), monte carlo simulation, probability of default, stochastic model, unexpected loss.

JEL classification code: C15; C63; G31; G32; G33.

1. Introduction

According to the classic analogy between corporate securities and options, a shareholder's payoff profile can be likened to that of a long call on the company's assets – generally defined as *Enterprise Value* (EV) – with strike price equal to the company's debt value; while the lender's payoff is likened to that of a short put option on the company's net asset value, with strike price equal to the net value of debt and expiration time equivalent to the maturity of the debt¹. In this analogy, shareholders, reimbursing the debt, pay the strike of the option and acquire the right to appropriate all of the net cash flows generated by the company; the equity value corresponds to the value of the call option and thus depends on the difference between EV and debt.

Based on this logic, a company can be considered in default when the overall value of its assets is inferior to the value of its debt. In fact, whenever the company's EV is inferior to the value of its debt, it is to shareholders' advantage not to reimburse the debt and relinquish company assets to creditors, as their possible losses are limited to the capital they have paid in². Therefore, from a theoretical point of view, a company's level of indebtedness becomes economically unsustainable whenever the overall value of company assets is inferior to the value of its debt.

The problem in credit risk analysis is verifying ex-ante the probability that such an event will occur in the future. An example of this logical schema is provided by the option/contingent models, such as the Merton model, derived from the options theory, in which the probability of default (PD) is established by the probability that the company's EV is inferior to the value of its debt³. The PD is thus a function of the debt, the EV and the EV's volatility. In the various practical modes of application of this model, debt is generally a constant, equal to the extant value at the moment of the analysis, while the other two key variables are estimated with reference to the market prices of the company's stocks.

Our purpose here is to present a stochastic model for estimating a company's PD based on the above-described relationship identifying the event of default, also used in option/contingent models. However, unlike the latter models, the proposed method leaves market values aside in estimating EV and its volatility, and removes the hypothesis of a static value of debt over time.

The RAPD (*risk analysis probability of default*) approach, which we intend to illustrate, directly measures the probability that default will occur, or the probability that in the future EV will fall below debt value. EV and debt are calculated endogenously within the model, in accordance with the theory of finance. Given the analytical complexity of the problem, the simplest way to obtain a stochastic model solution is to use Monte Carlo simulation techniques.

We will begin to describe the relationships that identify the condition of default, the logic on the basis of which the reconstruction of default scenarios occurs and how PD is determined, explaining why the simulation is in effect necessary to obtain the model solution. Next, we will illustrate how this approach also allows us to obtain a direct estimation of LGD (*loss given default*); successively, considering the common theoretical basis shared by the RAPD approach and option/contingent models, we will compare the two methods and highlight the main differences between them.

This paper's aim is to present a new methodological approach to corporate credit risk estimation, thus a full and exhaustive empirical analysis of the validity of the RAPD Approach goes beyond the scope of this research. Nevertheless, we also present an application of the method, with two purposes in mind. Firstly, we wish to offer an example of how the method can be implemented and replicated, keeping in mind that the "implementation hypothesis" adopted here must be considered as just one of the simplest ways to apply the method, and certainly not as the only or best way (better and more complex applications can – and we think should – be considered by professional analysts who are interested in using the

¹ As is known, this type of correspondence was first suggested by Fischer Black, Myron Scholes and Robert Merton in several articles published in 1973. See Black and Scholes (1973) and Merton (1973).

² Of course this theoretical statement is a simplification of what occurs in the real world, since it is management that runs a firm and not its shareholders; in any case, the statement conveys the idea.

³ This type of model is derived from Merton's initial contribution (1974); subsequently, a series of models was developed, successively removing some simplifying hypotheses of the original model. See for example Black and Cox (1976), Geske (1977), Vasicek (1984), Litterman and Iben (1991), Kim, Ramaswamy and Sundaresan (1993), Longstaff and Schwartz (1995), Jarrow, Lando and Turnbull (1997), Duffie and Singleton (1999).

RAPD Approach). Our second purpose is to provide a first look at evidence of the method's soundness, showing the results in a significant number of real cases.

We thus conclude with empirical evidence derived from application of the RAPD approach. The results of a series of back-testing analyses comparing various credit risk models carried out through ex-post analysis on defaulted and performing companies will be presented. Although the exercise must be considered strictly as a preliminary test, the results show how, through a replicable application of the RAPD approach (even in very simplified ways), one can obtain accurate PD estimates capable of satisfactorily describing the evolution of a company's credit risk, and of providing early warning risk indications that are stronger and ahead of the curve compared to PD estimates obtained with the other models considered.

2. RAPD Approach: a reduced formulation

Given the customary practice of considering annual periods as the interval of reference for PD estimation, we will place ourselves directly into a discrete set, in which each period is one year.

Suppose that the company's performance is conditioned by some elements of uncertainty, the dynamics of which can be described through a series of stochastic processes with discrete parameters, of which we assume that the information necessary to define them is known.

The following are the main notations of the symbols used:

$REV =$	Revenue.	$NIC =$	Net Invested Capital.
$g =$	Revenue Growth Rate.	$\tau =$	Corporate Tax Rate.
$m =$	Operating Margin (i.e. Ebit Margins).	$D =$	Debt.
$NOPAT =$	Net Operating Profit After Tax.	$r_d =$	Cost of Debt.
$NFA =$	Net Fixed Asset.	$INT =$	Interest Expenses.
$f =$	NFA/REV (Capital Intensity Factor).	$CASH =$	Excess Cash.
$NWC =$	Net Working Capital.	$\Delta CAP =$	Issue Additional Capital
$w =$	NWC/REV	$OCF =$	Operating Cash Flow

Let us now define a few fundamental relationships representative of the operating performance of a company over time, based on the variables defined above. First of all, let us define the expected company's $NOPAT$ and NIC , as:

$$[1] \quad NOPAT_t = REV_{t-1} \cdot (1 + g_t) \cdot m_t \cdot (1 - \tau)$$

$$[2] \quad NIC_t = (f_t + w_t) \cdot REV_{t-1} \cdot (1 + g_t)$$

The expected operating cash flow (OCF) at the time period t can be defined as:

$$[3] \quad OCF_t = NOPAT_t - \Delta NIC_t + \tau \cdot INT_t$$

The OCF represents the net cash flow generated by the company's operating business; it is a measure of the available cash flow claimed by all the capital investors in the company (shareholders and debt holders) that explicitly includes the effects of the tax shield of the debt⁴.

For the sake of simplicity, let us suppose that interest expenses, given the discrete set under consideration, are defined based on the standard hypothesis that considers an average stock of debt equal to the mean value between debt at the beginning and end of the period, or:

$$[4] \quad INT_t = r_d \frac{D_{t-1} + D_t}{2}$$

The variables at the basis of the calculation of the relations defined in [1], [2], [3] and [4] determine the company's economic performance and contain the elements of uncertainty that characterize it. The stochastic processes on which the model is based must therefore be specified considering these variables as stochastic, and thus defining their probability distribution functions and interdependence relationships. However, we will not formulate a specific modeling of these variables at this time, as, in practical application of the model, they may be defined in a variety of more or less complex and articulated ways. For

⁴ In technical jargon, this cash flow is denominated *capital cash flow*. See Ruback (2002). Further along (Footnote 10), it will become clear why this particular formulation of cash flows is utilized.

example, they may be defined by means of functions, linked through economic relationships, in which the stochastic component is reflected in only one of the variables or parameters of the function.

2.1. Financing policy and debt estimation

The first fundamental relationship on which RAPD is based is that which determines the stock of debt at the end of the time period t ; this variable is determined in such a way that it always fulfills a condition of perfect financial equilibrium for the company – that is, parity between cash inflows and outflows. To model this relationship, we must define debt based on a recursive process, so that its value at the end of each period is determined respecting a financial equilibrium constraint, which is a function not only of what occurred over the course of the period, but also of everything that occurred in preceding periods.

In practice, given the financial equilibrium constraint between cash inflows and cash outflows, the debt at the end of time period t will be a function of the stock of debt at the end of the preceding period, the OCF generated in the time period t and the total interest expenses connected to the debt itself. For the sake of simplicity, let us assume that there is no interest income on excess cash and no dividends distribution; we will remove these simplifications later on. Therefore, we can define the stock of debt (D) at the end of the time period t as:

$$[5] \quad D_t = \max[0, D_{t-1} - OCF_t + INT_t - \Delta CAP_t]$$

Equation [5] expresses a pure financial equilibrium constraint, in which the company's financing policy rules are implicitly set. The value of the debt can never be negative, but at best can be equal to zero in the case it is fully refunded.

To manage cases in which the company is cash positive ($D=0$), it is necessary to define a variable that expresses the excess cash value in each period. Considering equation [5] we can express excess cash as:

$$[6] \quad CASH_t = D_t - D_{t-1} + OCF_t - INT_t + \Delta CAP_t$$

Swapping equations [3] and [4] into equation [5] produces a simple recursive solution for the value of the stock of debt at the end of the time period t :

$$[7] \quad D_t = \max \left[0, \frac{2 \cdot (NOPAT_t - \Delta NIC_t + \Delta CAP_t - 2D_{t-1})}{r_d(1-\tau) - 2} - D_{t-1} \right]$$

Defining debt as the result of a financing policy determined on the basis of a recursive process – in which the debt depends first and foremost on the company's financial condition – rather than considering it in exogenous terms as a static variable, is in itself a strong element of complexity and intrinsic unpredictability in the model. But on closer inspection, this hypothesis is absolutely consistent with what actually occurs in a company, where debt is never a completely exogenous variable, and tends to be increasingly less so as financial conditions worsen – that is, when recourse to debt becomes more a necessity than the result of a deliberate management choice⁵.

In any case, we must keep in mind that this assumption does not preclude us from considering – while continuing to observe the constraint of financial equilibrium – financing policies that follow exogenous rules, such as, for example, a pre-determined debt repayment plan, maintaining a certain set relationship between debt and equity, or rules that call for observance of particular covenants⁶. In general, more or less complex structures can be utilized in defining the model, for financing policy and dividend policies, making recourse to appropriate functional relationships and control variables.

⁵ We are simply assuming that companies tend to issue debt whenever self-financing is insufficient to cover their financial needs. This assumption can be traced back to the so-called *pecking order theory*, which hypothesizes that companies' financing choices are not always made in pursuit of a fixed debt/equity target ratio, but vary with changing circumstances. More specifically, it asserts that companies make recourse to external financing only when internal funds are insufficient. Furthermore, in choosing the system of external financing, a company prefers to utilize debt or debt-like securities, and only issues equity as a last resort. See Myers and Majluf (1984) and Myers (1984). This type of explanation also seems to be confirmed by some empirical analyses; see in particular Graham and Harvey (2001).

⁶ For example, optimal financing rules that tend to maximize net value at any given moment could be structured. On these aspects see in particular Fischer, Heinkel and Zechner (1989), Leland (1994), Goldstein, Ju and Leland (2001), Dang and Zechner (2004), Tirole (2006).

For example we could set a simple rule that ensures that the value of debt does not fall below a certain predetermined minimum level (which may represent a target leverage ratio), just by changing equation [5] in the following way:

$$[8] \quad D_t = \max[\bar{D}, D_{t-1} - OCF_t + INT_t - \Delta CAP_t]$$

Where $\bar{D}$ is the minimum level of Debt.

Dividend policy⁷ could be easily included in the model by setting a rule through which in each period all or part of excess cash available is distributed as dividends, or more simply by setting a specific dividend payout parameter d . In the latter case, equation [5] changes to:

$$[9] \quad D_t = \max[0, D_{t-1} - OCF_t + INT_t - \Delta CAP_t + d_t \cdot (NOPAT_t - INT_t \cdot (1 - \tau))]$$

As a consequence, equation [6] will be:

$$[10] \quad CASH_t = D_t - D_{t-1} + OCF_t - INT_t + \Delta CAP_t - d_t \cdot (NOPAT_t - INT_t \cdot (1 - \tau))$$

It follows that, swapping equations [3] and [4] into equation [9], the stock of Debt at the end of period t will be:

$$[11] \quad D_t = \max\left[0, \frac{2 \cdot (NOPAT_t \cdot (1 - d_t) - \Delta NIC_t + \Delta CAP_t - 2D_{t-1})}{r_d \cdot (d - 1) \cdot (\tau - 1) - 2} - D_{t-1}\right]$$

The relationship structure of the model can be further developed and complicated on the basis of even more general conditions, without substantially changing the logic of the model solution. For example, we could consider the following complications to make the model more suited to describe real conditions:

1. Excess cash and financial investments generate interest income.
2. Depreciation can be calculated explicitly through an amortization plan; this allows for accurate estimation of the effective expected capital expenditures, or Capex, connected to a preestablished level of net fixed asset determined as a function of the NFA/REV ratio⁸.
3. Debt structure and its cost can be split into long-term debt and short-term debt, structuring the model so as to determine the value of long-term debt to cover the financial need arising from Capex and short-term debt to cover the Net Working Capital financial need.

These three models' complications were adopted in the model used for the empirical analysis reported later on.

⁷ In any case, consider that for PD estimate purposes, dividend policy is really relevant only for PD referring to time horizons longer than one year.

⁸ Capex can be obtained considering that the NFA at time t can also be expressed as:

$$[I] \quad NFA_t = NFA_{t-1} + CAPEX_t - DA_t$$

where DA is the value of depreciation and amortization, which, assuming that Capex is gradually carried out through the year, can be defined as:

$$[II] \quad DA_t = \alpha_t \left(GFA_{t-1} - RETIR_t + \frac{CAPEX_t}{2} \right)$$

where α is the expected rate of depreciation and amortization, GFA are the gross fixed assets, $RETIR$ is the value of retirements (i.e. the share of fixed assets fully depreciated that must be deducted from GFA every year). Swapping equation [II] into [I] we can obtain the Capex as:

$$[III] \quad CAPEX_t = \frac{2 \cdot (NFA_t - NFA_{t-1} + \alpha_t \cdot GFA_{t-1} - \alpha_t \cdot RETIR_t)}{2 - \alpha_t}$$

$RETIR$ can be determined in several different ways. One way is through a closed formula; for example assuming that NFA in each period is equal to the sum of all the future depreciations necessary to fully depreciate the outstanding GFA :

$$[IV] \quad NFA_t = \sum_{i=1}^n \alpha \cdot \left(GFA_t - i \cdot \frac{GFA_t}{n} \right)$$

And resolving equation [IV] in order to obtain n , we can then determine $RETIR$ as GFA divided by the number of years necessary to its full depreciation:

$$[V] \quad RETIR_t = \frac{GFA_t}{n}$$

Another way to determine $RETIR$ is to multiply the retirement rate (r) by the GFA of the previous period, recurring to an historical estimate of the retirement rate:

$$[VI] \quad RETIR_t = GFA_{t-1} \cdot r_t$$

Considering that, in the absence of disinvestments, retirements in each period are equal to the value of depreciation and amortization minus the change in accumulated depreciation, estimates of r can be obtained dividing the historical values of this difference by GFA values.

2.2. EV estimation

The second fundamental relationship on which RAPD is based is the definition of the enterprise value (EV) at the end of the time period t . Given the hypotheses expressed above, this can be expressed in general terms as:

$$[12] \quad EV_t = \sum_{i=1}^{T-1} \frac{OCF_{t+i}}{(1+k)^{i+1}}$$

in which k represents the discounting rate. In [12], for the sake of simplicity we implicitly assume that the company is liquidated at the end of period T . However, in concrete application of the model, it is generally opportune to consider a further component of value provided by the terminal value⁹.

Assuming that cash flows related to every possible scenario are expected cash flows and not risk adjusted cash flows¹⁰, and with the intention of estimating the risk of default – defined as the probability that debt becomes greater than the effective market value of company assets (or the theoretical value at which expected cash flows would be exchanged in a competitive capital market), – we must discount back the cash flows at a risk-adjusted rate, implicit in which is an adequate risk reward for both lender remuneration (i.e. interest expenses) and shareholders. Given these assumptions, the discount rate to be used is the *weighted average cost of capital* ($wacc$)¹¹. The underlying hypothesis is that shareholders gain from financing the company up to its market value, and trigger its default when they realize that the market value of the company's assets is inferior to that of its debt.

2.3. Event of default and PD estimation

Equation [12] not only defines the value of the company's assets, but also individuates what we can define as its maximum debt capacity. In fact, any level of debt above said threshold cannot be reimbursed by means of cash flows that will be generated by company assets. Therefore, the company's theoretical maximum debt finds its natural, insurmountable limit in the EV. In such an extreme case, the totality of the OCFs generated by the company would go to reimburse lenders, leaving no residual value for shareholders; equity value would thus be at zero.

Considering the possibility that the company might have an excess cash as well, we can easily define the event of default. In fact, a company would find itself technically in a condition of default whenever:

$$[13] \quad EV_t < D_t - CASH_t$$

The purpose of the model is to estimate the expected frequency with which the event of default as defined in [13] occurs; that is, the probability that EV will drop below the value of net debt¹².

⁹ We suggest using as a terminal value a simple perpetuity formula with no growth considering *NOPAT* as proxy of the terminal cash flow, since within the stochastic framework proposed, in the worst-case scenarios generated in the simulation (which are those more relevant for the calculation of the probability of default), company value can be reasonably expressed by the value of cash flows from assets in place as a perpetuity with no growth, under the assumption that a buyer will not pay for future investments in conditions of distress.

¹⁰ See Damodaran (2008), pp. 177-178.

¹¹ Naturally, in practical terms the precise configuration of the cost of capital to be used depends on how cash flows are defined. We must keep in mind that, in attempting to obtain a correct estimation of a company's EV, we have the problem of correctly estimating in every period the tax effects on value connected to the specific financial structure that is dynamically created as a result of balancing on the debt side. The simplest and most direct way to obtain coherent estimations is to use valuation models such as APV (*adjusted present value*), or other models derived from it, such as *capital cash flow*; the latter is the methodology we are implicitly utilizing in the specification of the model and this is the reason why we used this cash flow formulation. For more on these subjects see for example Myers (1974), Kaplan and Ruback (1995), Gilson, Hotchkiss and Ruback (2000), Ruback (2002), Fernández (2002), in particular, pp. 443-470, Fernández (2004), Kruschwitz and Löffler (2006).

¹² In a less simplified formulation of the model, EV could be adjusted to take into account a series of factors, so as to provide a debt reimbursement capacity estimation that is closer to the actual situation under analysis. These may include in particular: bankruptcy costs (if they have not been considered implicitly in the discounting rate) and pension and post-retirement liabilities, which include long-term provisions for pension and post-retirement benefits that are underfunded. We may also consider, in all future scenarios in which the expected value of the company is inferior to already granted long-term debt, an EV at least equal to

Every possible pair of debt and EV values can be imagined as the realization of a joint stochastic process, on the basis of which every possible value of debt, at the end of every period, is placed in relation with “all” possible values of EV that could correspond to that value of debt. Given that by definition PD is linked to the company’s overall risk, analysis of the condition of default can be imagined as the coming to pass of every possible alternative joint event, represented by the comparison of two sides of the financial situation: the stock of debt accumulated up to that moment and the total cash flow available to repay debt that will be generated from that moment on. This system of conditioned verification also allows us to ensure financial coherence for the entire process, in that it respects a necessary relationship of dependence that requires that the company’s financial conditions at the end of a given period (synthesized in the stock of debt accumulated) depend on the economic-financial evolution of the company up to that moment; while future repayment capacity (represented by cash flows that will be generated from that moment on) naturally depend on what will happen in the future, which is in turn conditioned by the company’s current situation, determined subsequent to what has occurred up to that moment. In this regard, summarizing the economics underlying the RAPD model, we may say that the firm’s value creation capacity, solvency and debt reimbursement capacity (or liquidity) are intrinsically tied to each other and all have the same generation process¹³.

Theoretically, the solution to the stochastic model could also be obtained by seeking an analytical solution based on the equations that define the model. However, within a more general context, the model would end up incorporating a large number of stochastic and state variables with a complex path-dependent structure. At that point, an analytical solution – presuming one could be found – would be too complex. Furthermore, an analytical solution would in any case be tied to the functional relationships of the model and the probability distributions utilized, hence any change in the model or probability distribution functions would require the individuation of a new solution.

The simplest way to solve the model, while maintaining a reasonable level of flexibility, is to utilize a Monte Carlo simulation process, obtained through a copula-based approach that allows expression of the joint distributions of random variables as a function of the marginal distributions and pairwise correlations to incorporate dependence among the variables¹⁴. The construction of a probabilistic model of this kind allows the user to make subjective judgments of marginal distributions and correlations and easily generate a large number of random scenarios, and check each one for the condition of default indicated by equation [13]¹⁵.

the value of the debt in which reimbursement is not immediately due, and therefore assume that EV is equal to the greater of the values between available long-term debt and the present value of cash flows after other adjustments. This type of set-up is equivalent to that used in several modelizations derived from option/contingent models. For example, in the Moody’s KVM model, based on an option/contingent approach, debt (the definition of which changes in different generations of the model) is determined as short-term debt plus half of long-term debt.

¹³ It is interesting to note that this kind of economic interdependence is also present in H. P. Minsky’s financial instability theory. In fact, in his theoretical framework the concepts of “solvency and liquidity constraints” are defined in exactly the same terms as in the RAPD model: «Given the valuation of assets and liabilities one may derive a net worth or owner’s equity for the unit. [...] A unit is solvent, given a set of valuation procedures, when its net worth is positive. [...] it would be better if valuation procedures were conditional, that is, of the form: if the economy behaves as follows, then these assets would be worth as follows. A unit is liquid when it can meet its payment commitments. Solvency and liquidity are two conditions that all private economic organizations must always satisfy. Failure to satisfy either condition, or even coming close to failing, can lead to actions by others that affect profoundly the status of the organization. Even though textbooks may consider solvency and liquidity as independent attributes, the two are interrelated. First of all, the willingness to hold the debt of any organization depends in part upon the protection to the debt holder embodied in the unit’s net worth. A decline in net worth – perhaps the result of revaluation of assets – can lead to a decreased willingness to hold debts of a unit and hence to difficulties when it needs to refinance a position. A lack of liquidity may result from what was initially a solvency problem». See Minsky (1972).

¹⁴ Cfr. Rubinstein (1981), Robert and Casella (1999).

¹⁵ An early attempt to develop an approach to credit risk that goes beyond traditional financial ratio analysis, introducing probability in assessing risk of default, is Wilcox’s Gambler Ruin Approach (see Wilcox 1971, 1976). However, this model, as the author himself admits, is not suited to estimate precise PD, but is rather a method to discriminate between high- and low-risk firms, for relative risk valuation (somewhat similarly to scoring models). Furthermore, the Gambler Ruin Approach differs from the RAPD Approach substantially with regard to: the definition of default based on the concept of net assets liquidation value (set to a zero liquid wealth, i.e., a firm defaults when it runs out of cash), and the use of static accounting data for determining the event of default [while the RAPD Approach applies stochastic and dynamic values and bases the definition of default on economic values (i.e. assets valued through DCF)]. Later, Schwartz and Moon (2000, 2001) outlined an approach similar to RAPD for PD estimation, but although it is in some ways analogous to that of Schwartz and Moon, the RAPD model differs

Once all cases of default occurring in the simulation have been individuated, we can easily determine the PD relative to each period as the relative frequency between the number of defaults occurring in the period and the total number of scenarios simulated. So, for example, the PD at one year is calculated by dividing the number of default events that occur in the first year by the total number of trials in the simulation. The PD at two years is determined by the relationship between the number of scenarios in which default occurs in at least one of the first two years and the total number of trials. Analogously, the PD at n years is determined by dividing the number of scenarios in which default occurs in at least one of the first n forecast years by the total number of trials in the simulation.

In formal terms, (assuming for simplicity that $CASH = 0$) we can define the cumulated PD on various time horizons in the following way:

$$\begin{aligned}
 PD_1 &= P(EV_1 < D_1) \\
 PD_2 &= P(EV_1 < D_1) + P(EV_2 < D_2 | EV_1 > D_1) \\
 [14] \quad &\dots\dots \\
 PD_n &= P(EV_1 < D_1) + P(EV_2 < D_2 | EV_1 > D_1) + \dots \\
 &\quad + P(EV_n < D_n | EV_1 > D_1, EV_2 > D_2, \dots, EV_{n-1} > D_{n-1})
 \end{aligned}$$

where each element of probability, the sum of which defines the PD in each year, can be defined as the marginal annual probability of default, or the probability that default will occur in that year, presuming that the company had not already defaulted in one of the previous years. More precisely, this calculation method allows us to individuate three types of probability:

1. *Yearly Default Frequency*: indicates the frequency of scenarios in which default occurred in that given year. In this sense, it provides a measure of the company's degree of financial fragility in that specific period. $[P(EV_t < D_t)]$
2. *Yearly Marginal Default Frequency*: indicates the frequency with which default occurs in a given year, but only in cases in which default has not already occurred in previous years. In this sense, it provides a measure of the increase in overall risk determined in that given year. $[P(EV_t < D_t | EV_1 > D_1, \dots, EV_{t-1} > D_{t-1})]$
3. *Cumulated Default Probability*: provides a measure of overall risk of default along a given time horizon, and is determined by the sum of the marginal default probabilities. $[P(EV_1 < D_1) + \dots + P(EV_t < D_t | EV_1 > D_1, \dots, EV_{t-1} > D_{t-1})]$

2.4. RAPD default scenarios

In order to better describe the logic and dynamics of the RAPD approach, it is useful to provide a graphic representation that helps us to imagine the process of reconstructing default scenarios, simulating – on the basis of the simplified model described above – 3,000 possible scenarios that describe the future evolution of a hypothetical company, with an initial stock of debt of 700¹⁶.

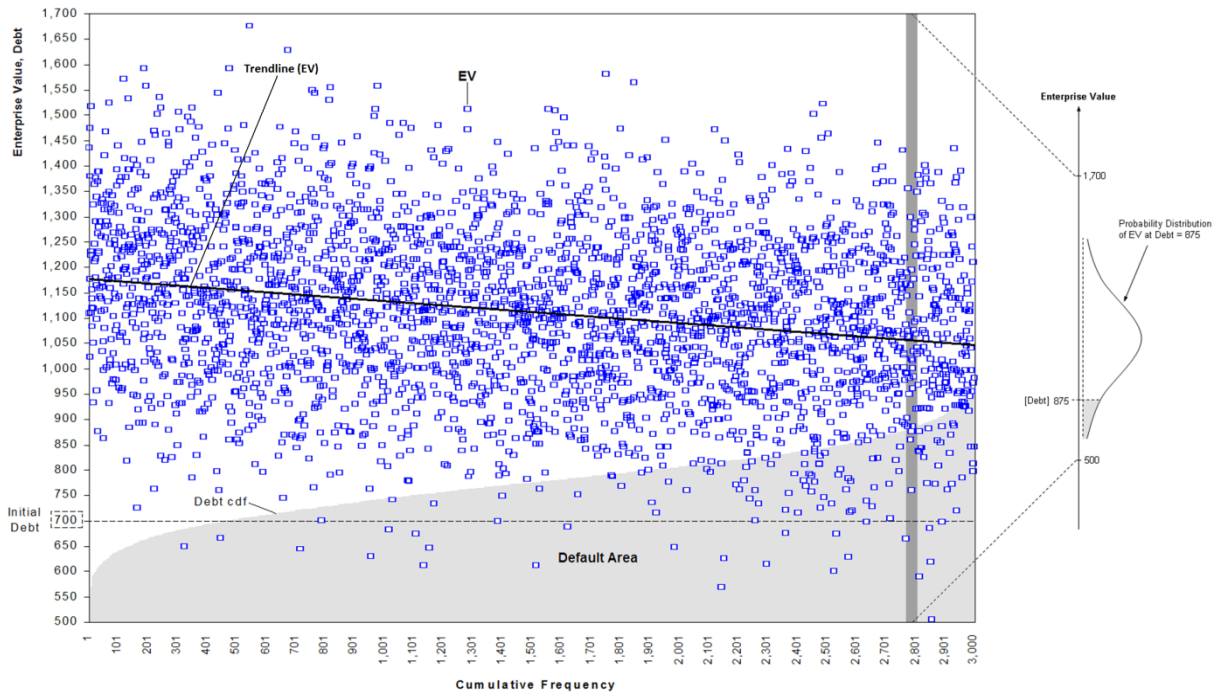
Fig. 1 reconstructs the cumulative distribution function of the possible debt values at the end of the first period (*Debt cdf*), which arranges in order from lowest to highest all of the debt values obtained in the 3,000 simulation trials. For each value of the Debt cdf, all of the corresponding EV values (the square markers), realized in the various scenarios simulated relative to that specific debt value, are indicated on the graph. If we imagine moving on a hypothetical vertical line through the diagram, shifting horizontally from left to right, we run through the possible values of the Debt cdf, from smallest to largest, and at the same time, along the vertical line, we can individuate the possible EVs linked to each specific debt value. In this way we can individuate all of the possible scenarios of default for the company, with the default points lying below the Debt cdf, which delimits an area in which $EV < D$.

from it substantially, in particular with regard to the definition of default and the estimation of variables on the basis of which the condition of default is set.

¹⁶ Consider that the graphic representation shown in Fig. 1, with all of the simplifications of this case, depicts the results of a real application of the RAPD approach.

For example, on the vertical line highlighted in grey, we can observe that several EVs correspond to the debt value of 875 on the cdf; these are in practice all of the EVs that emerge in the simulated scenarios contemporaneously with the debt value of 875. If we increase considerably the number of trials of the simulation and magnify the portion of the graphic highlighted by the gray line, we can see that many possible EVs correspond to a debt value of 875, which can in turn be arranged in the form of a distribution function, highlighted on the right side of the diagram¹⁷. The probability of default corresponding to a debt of 875 [$P(EV < 875)$] is given by the area of distribution of EV highlighted in gray, to the right of the value of 875. The same reasoning can be extended to all the debt values on the Debt cdf. Hence, the PD value will depend on the total number of EVs inferior to the corresponding debt values, or all of the EVs that lie within the default area highlighted in gray below the Debt cdf.

Fig. 1 - RAPD default scenarios: Debt & EV joint correspondence



Finally, it should be pointed out that, as far as there exists a certain time dependence in the company's economic evolution; and as far as it comes to a condition in which debt no longer carries any tax benefit, we must expect that lower EVs will tend to correspond to higher levels of debt. The case represented in Fig. 1 exemplifies just such a situation. In fact, as we can see, the trendline, sloping negatively, shows that on average, higher debt values tend to correspond to lower EVs. This means that in those scenarios in which debt tends to increase without carrying any increase in future returns, EV cannot help but decrease. In these cases, therefore, peaks in debt level will tend to be associated with lower EV; defaults will thus tend to be concentrated for the most part in this zone. This relationship is effectively captured by the RAPD, in that debt and EV are determined endogenously and jointly by the model. Fig. 1 also clearly shows what the PD error would be considering a constant level of debt, in our example set at an initial debt of 700. In fact, we can see that the error in estimation of PD, considering a constant debt, is given by the number of default points present in the section of the default area between the Debt cdf and the line corresponding to a constant debt value. As we can see, any hypothesis of constancy of debt would entail more or less significant distortions.

¹⁷ The image of the distribution function is only a theoretical illustration.

3. Estimate of loss given default (LGD), expected loss (EL) and unexpected loss (UL)

As we have shown, with the RAPD approach we can identify all of the cases in which default occurs as defined in [13], precisely calculating both debt value and EV for each scenario simulated. Due to the way in which they are calculated within the framework of the RAPD, these two variables can be utilized not only to individuate the event of default (and therefore PD), but also to estimate *loss given default* (LGD) and thus implicitly the debt *recovery rate*.

In order to calculate LGD, we need to know the company's exposure at the moment of default (*exposure at default* – EAD) and the value of the debt recovery in case of default. Now, let us consider that the event of default occurs when, in practice, the financial market (banks, bond holders, etc.) does not fully cover the company's financial needs; and also consider that the debt value determined by the model described represents the company's total financial needs at the moment of default, whatever the maximum value of debt that the financial market is willing to lend may be (in the context of this model, it is represented by EV). In fact, the financial equilibrium constraint implicit in equation [5] ensures that the company's debt at the end of each period always allows for a perfect balancing between cash inflows and outflows, thus considering all of the company's liabilities, not only those relative to the financial market, but also with regard to all of the various other parties, i.e. employees, suppliers, government, etc. Hence any part of debt not covered by the financial market should in any case be considered as outstanding with regard to the other subjects.

Therefore, the debt determined by the model at the moment of default, effectively represents the company's total EAD, and so within the model we will have $EAD = D$. At the same time, the EV in a given scenario constitutes – at least in theory – the best ex-ante estimation of the post-default recovery value of the company's assets in that scenario. Therefore, EV can be considered as the presumable recovery value of the company's assets post-default, and constitutes the second element necessary for the calculation of LGD.

At this point, if we consider only the simulation scenarios in which default occurs, and assuming for the sake of simplicity that there are no differences in seniority rights between the various holders of the company's liabilities, the LGD at period t , for the k -th default scenario can easily be calculated based on the following expression¹⁸:

$$[15] \quad LGD_t^k = EAD_t^k - EV_t^k - CASH_t^k$$

It is important to note that with the RAPD approach the LGD can be calculated in a way consistent with the time horizon relevant to the analysis; in fact, depending on the number of forecast years considered in the model, we can obtain an estimated LGD for every period within the forecast time horizon (LGD at 1 year, at 2 years, etc.). Furthermore, calculating the LGD on the basis of this methodology, the PD (i.e. the event of default) and the LGD would prove to be consistently linked, so as to capture the relationship of interdependence that exists between the two variables¹⁹.

Once LGD has been calculated for all of the default scenarios in the simulation, it is possible to re-construct the distribution function of the LGD specific to the company. This allows us to calculate the *expected loss* (EL) by means of the average LGD value²⁰. If we denote with S the number of default scenarios, the average LGD can be defined as:

$$[16] \quad \text{Average}(LGD)_t = \frac{\sum_{k=1}^S EAD_t^k - EV_t^k - CASH_t^k}{S}$$

¹⁸ We should specify that the estimated LGD obtained thusly refers to *unsecured LGD*, or LGD related to the repayment of exposure in the absence of guarantees and collateral. Further on, we will illustrate how it is possible to consider the effect of guarantees and of different levels of debt seniority.

¹⁹ The recovery rate is therefore an endogenous variable, and is a function of the residual value of the defaulted company's assets.

²⁰ In the context of traditional credit risk models, EL is generally calculated on the basis of the following expression:

$$EL = PD \times LGD \times EAD$$

in which LGD is expressed as an average rate of loss in the case of default; in our case, we could make reference to the average value of the distribution function of LGD.

For the calculation of *unexpected loss* (UL), we need only directly determine, by means of calculation of percentiles, the LGD relative to the confidence interval of interest.

Through this method, then, we can reconstruct the distribution function of the LGD specific to the company in question, making it possible to directly and specifically estimate the relevant EL and UL values for each particular case, choosing the desired confidence interval and time horizon.

The schema we have just outlined can be generalized to consider calculation of LGD in a situation in which there are different levels of seniority and/or the presence of collateral and guarantees. For this purpose, we need only complicate the calculation a bit, considering the different priorities for repaying holders of the various installments of the financial debt. In practice, calculation of the LGD, for each class of creditors, can be effectuated by inserting adequate conditions into the model, so that EV is first utilized to repay those with higher seniority rights, determining an LGD only in the case in which EV is not sufficient to cover that part of the company's liabilities; subsequently, the residual EV can be used to repay the other classes of creditors, and the relative LGDs can be determined.

At the same time, we can complicate the picture with the presence of collateral and guarantees granted to the company; to take these into account, we need only add the value of the collateral and guarantees to the EV.

We estimated LGDs and recovery rates with the RAPD approach for a sample of defaulted companies²¹; results are reported in Appendix B. The estimates provide results in line with the empirical estimates provided by several studies²², with a sample average recovery rate of 37.96%, a median recovery rate of 33.65% and a standard deviation of 25.01%. In the simulation, we did not consider any collateral or guarantees, therefore recovery rates may be underestimated.

Recovery rates also show a negative correlation with PDs (Pearson's correlation = -0.704; Spearman's rank correlation = -0.728) as suggested by empirical research²³, confirming that the higher the PD, the higher the LGD. This relationship is structurally intrinsic to the RAPD model for two reasons. The first reason is that EV is negatively correlated with PD (the higher the EVs generated in the simulation, the lower the frequency of the default event), and the second reason is that in the worst scenarios characterized by low EV, Debt also tends to increase, thus determining a lower recovery rate (and a positive correlation between EV and recovery rate); this latter relationship is better explained in paragraph 4 below.

4. RAPD approach vs. option/contingent models

As we have already said, both models are based on the same economic relationship identifying the event of default [13]. Aside from this, the two approaches, albeit based on very similar logics, differ both in terms of the way in which EV and its possible future values are determined, and in the way debt is considered. Obviously, these differences also give rise to a different way of configuring the reconstruction of default scenarios and consequently the system utilized to calculate PD.

In option/contingent models, EV is calculated depending on current market prices, and its volatility is determined on the basis of the historic evolution of market prices. Debt is generally a static number, pre-determined independently of the model, or evolves according to some deterministic rule, in any case not linked to EV²⁴.

In the RAPD model, on the other hand, everything is based on the simulated evolution of company fundamentals, that is, on the economic-financial variables on which its financial requirements depend, which in turn determines the stock of debt. The operational evolution and financing policy management

²¹ For more detail about the sample and the assumptions adopted for the RAPD model application see section 5 "RAPD Approach: credit risk model comparative empirical test".

²² For example, Altman, Brady, Resti and Sironi (2005) report a 37.2% weighted average recovery rate over the period 1982-2001; Altman and Hotchkiss (2006), p. 324, report a 34.44% average (40.67% median) recovery rate with a standard deviation of 24.97% over the period 1974-2004.

²³ See Altman, Resti and Sironi (2005), Altman, Brady, Resti and Sironi (2005), Altman and Hotchkiss (2006), in particular p. 326-327.

²⁴ Not coincidentally, practical implementation of these models often requires ad hoc hypotheses relative to the debt itself. Many studies show that the individuation of the threshold debt value in these models does not always have a clear and precise theoretical justification. See Ong (1999) and Saunders (1999).

then jointly determine the EV for each period. Therefore the two variables relevant for estimating PD, stock of debt and EV, are simultaneously determined by the model in an endogenous and coherent way, on the basis of shared conditions and specific relationships that link them to one another, in accordance with well-known principles of finance theory. Unlike in option/contingent models, only EV varies according to its own probability distribution, while debt is a datum substantially exogenous to the model.

This is an important difference that demonstrates its relevance especially when it is necessary to estimate PDs that go beyond the short-term (1 year); for these cases it would be extremely limiting to consider debt as an exogenous and static datum, independent of the evolution of EV over time and the financial needs of the company²⁵. In practice, the risk is that of effectuating non-significant comparisons over time between high and low EVs and a debt value that remains steady or in any case is not linked to EV.

At this point, we must consider that both economic theory and empirical evidence clearly indicate that adjustments on the debt side normally tend to be asymmetrical; that is, debt may tend to increase when the company increases in value, but does not tend to decrease – in fact, it increases – when the company decreases in value. In the former case, the company has a clear potential advantage in increasing debt (linked to the tax benefits thus generated), and in the latter, it would have no potential advantage in maintaining or increasing its level of debt, but financial conditions do not allow for its reduction.

Disregarding these complexities, which are deeply rooted in the real situations that are the object of our analyses, can lead to a significant underestimation of PD value, especially in all of those cases in which companies have consistent debt. The graphic representation of PD estimation error in a hypothesis of constancy of debt, shown in Fig. 1, clearly illustrates this phenomenon.

It should be highlighted, then, that in option/contingent models, PD values are closely linked to market price evolution, and thus they incorporate an effective increase in PD only when there is a sharp drop in prices. Furthermore, the validity of the results of these models is highly dependent on the “quality” of the prices expressed by the market. Efficient markets will determine risk estimates coherent with available information and the fundamental values expressed by companies, while erratic market price evolutions will inevitably result in erroneous risk valuations. Naturally, this is not so much a theoretical limitation of the model as of the markets, which nonetheless operationally reflect on estimates produced by the model itself. Both theoretical reasons and empirical evidence lead to the exclusion of the hypothesis of perfect market efficiency. Recurrent financial bubbles and stock market crashes, as well as less conspicuous but more common phenomena such as price overreaction to announcements, clearly indicate how phenomena of market inefficiency are quite frequent and significant, both with regard to individual stocks and in terms of the overall market. In this context, option/contingent models, based in a “mechanical” way on market data, risk providing distorted and misleading PD estimates.

Finally, the option/contingent models’ need to array market prices as input data, appreciably limits their sphere of operational utilization, in that it keeps such models from being directly applicable to all private companies, at least not without making recourse to extremely restrictive hypotheses that would further weaken the reliability of results²⁶.

In any case, regardless of this, it is necessary in our opinion to have company credit risk estimates that are not simply the result of a calculation implicit in certain market values, which risk making the process circular and self-referential: market data $\Rightarrow$ implicit risk valuations $\Rightarrow$ market data. The use of valuation models and techniques that base their results on observations from the market itself, and not on fundamentals on which market values should depend, risk triggering a dangerous process of circularity in which the market ends up justifying and supporting values that it itself expresses, regardless of their economic coherence and/or sustainability.

²⁵ For some evidence regarding the lack of usefulness of option/contingent models for time horizons longer than two years see Miller (1998).

²⁶ I.e., considering EV and its volatility directly linked to the evolution of comparable companies, such as benchmark stock indexes for the sector.

5. RAPD approach: credit risk model comparative empirical test

Although the main purpose of this paper is to present and illustrate the modeling features of a new approach to corporate credit risk assessment, we also provide some preliminary empirical evidence on application of the RAPD approach, with a dual aim: to show how to apply the methodology, and to provide initial back-testing results. Consider, however, that a test of the RAPD approach is necessarily a joint test of the model outlined above [i.e. the forecasting model and the stochastic approach to PD calculation] and the specific set of hypotheses assumed to implement the approach [i.e. the choice of stochastic variables and their probability functions, setting of probability function parameters values and their evolution over time, setting of correlation parameters, etc.]. Therefore, the results necessarily depends on the soundness of both the methodology adopted and the assumptions made in its implementation. As better clarified further on, in the back-testing analysis we applied a very simple set of hypothesis in implementing the method; better results can be obtained with the use of more information and data.

In back-testing the RAPD model, we applied the method to both companies that later defaulted (defaulted companies) and companies that were still performing at the date of the analysis (performing companies). Analyses were carried out for several consecutive years, so as to verify RAPD's capacity to measure the evolution over time of companies' credit-worthiness, and the results provided by RAPD were compared with credit risk estimates derived from a few well-known and widely utilized measures of credit risk: Altman's Z-Score model²⁷, Moody's KMV's EDFTM model, Bloomberg Credit Default Risk (function DRSK)²⁸, CDS (Credit Default Swap) and the evolution of ratings issued by Standard & Poor's.

The comparative back-testing analysis, although based on a significant sample of companies, must be considered only as a preliminary test, and has a dual purpose. Firstly, it is intended to test RAPD's effective credit risk measurement capacity jointly with specific implementable paradigm hypotheses adopted for the development of the model and of stochastic simulations. Secondly, it seeks to verify which of the various measures (each representative of a different class of credit risk models) produces the best results; and more specifically for the sample of defaulted companies, which model is capable of best predicting the increase in credit risk that successively culminates in default.

Before moving on to illustrate the results, we should mention a few caveats of the analysis, in order to clarify the limitations and problems inherent in this type of empirical test:

- To compare the various credit risk models, we must first and foremost establish a common unit of measurement on the basis of which to compare results. The natural basis for this type of comparison is obviously PD. But unfortunately, only RAPD, Moody's KMV and Bloomberg DRSK models provide direct estimates of PD. Rating agencies provide risk measurements in alphanumerical terms, while the Z-Score provides only a score useful for putting subjects' risk levels into cardinal order, but not for expressing an absolute measure of each one's degree of risk. It was thus necessary for the purposes of comparison to translate ratings and scores into PDs, using a reconversion scale (master scale). Z-Score had to be first converted into rating equivalent and then into PD. CDS are transformed into PDs using a widespread formula²⁹.
- The Moody's KMV model limits estimated PD to a maximum value of 20%, extended to 35% in the "EDF 8.0 Model" released in 2007. This threshold value makes it impossible to precisely quantify the difference in the measurement of risk estimated in the worst cases, those on the verge of default. In any case, within the sample considered in the analysis this limit is not a major issue, since the PD cap applies only in 17 default cases (12 cases related to 1-Yr before default; 3 cases 2-Yr before default; 1 case 3-Yr before default) and none of the non-defaulted cases.
- While the results relating to comparative back-testing analyses of default cases allow for simpler and less ambiguous verification of the performance of various credit risk models, comparison of the results of "performing companies" is indubitably more uncertain. In fact, in the default cases interpreta-

²⁷ See Altman (1968).

²⁸ Moody's KMV and Bloomberg DRSK are two models based on the Merton distance-to-default. For more information about their specific technicalities and assumptions see respectively: Vasicek (1984); Singenellore, Lee, Li, Mann and Rajat (2012).

²⁹ For more details about the methodologies adopted to compare the different credit risk models results and the caveat of this kind of comparison see the Appendix A.

tion of the models' results is clearer, since the model that first and most markedly indicates increases in risk is also the one that is able to best predict successive default. However, for "performing companies" there is no objective criteria to state which PD value is the most appropriate to measure the company's degree of risk (or in other words, the company's *true* credit risk), making for more ambiguous and subjective interpretations of the results. In any case, at least two important checks can be done in back-testing performing companies. The first is to check whether the RAPD approach presents a systematic bias, upward or downward, in terms of PD estimates compared to the other models. The second is to check how the RAPD approach performs in measuring very high-credit standing companies (i.e. AAA stable over time) for which any sound model should reasonably provide an extremely low PD. Similar consideration can be made for companies with a very high credit risk profile (i.e. CCC or lower rating). Otherwise, the most controversial cases to compare are those involving companies with a credit risk around investment grade, whose credit worthiness is inevitably more subjective.

- One of the most important aspects of the RAPD approach is that it allows for (and at the same time presupposes) the utilization of all relevant information available at the time of analysis for the development of the simulation. Thus, any available information can be utilized to make inferences on PD estimates. However, in back-testing analyses carried out ex-post, we were unable to fully render this type of conception of the RAPD model, as it was necessary to apply extremely simplified standard implementation paradigms based on a set of information pared down to the strictly essential. Consequently, for example, no expected evolution in a company's financial dynamics was taken into account in the simulations, nor was information regarding capital increases, announced restructuring plans and other relevant expected events that may have been widely known about in the period under analysis. The PD estimates obtained thus do not necessarily represent those that an adequately informed analyst could have reckoned if he had applied the method to individual cases at the time to which the analysis refers. Of course, this aspect may negatively bias the estimates obtained in this test with the RAPD approach, in that said information could have been accounted for, more or less correctly, in models based on market prices (Moody's KMV, Bloomberg DRSK and CDS) and in judgments made at the time by rating agencies. In particular, in cases regarding companies that were passing through periods of great difficulty with negative financial performances, the application of the RAPD method with simplified implementation paradigms that do not account for expectations regarding possible new risk-mitigating elements may provide PDs significantly higher than those that would be estimated if such elements were included in the simulation. The decision not to consider a set of information that would certainly have been available at the time of initial analysis depends simply on the difficulty of obtaining such information years later; and in any case, the ex-post use of that information could have affected the fairness of the comparative analysis performed.

5.1. RAPD comparative back-testing analysis: methodological aspects

The back-testing analysis considered 46 cases of company default – some very well known – occurring in about the last ten years. The number of cases is limited to those for which all data required for the analysis were available. The analysis was carried out using the various models to estimate PD at one year, two years and three years prior to default, thus three PD estimates were collected for each case³⁰. The need to utilize companies that defaulted, were listed and had track records of ratings significantly restricted the number of possible cases. Furthermore, we excluded companies with extremely low ratings, i.e. those for which near-default conditions had already been acknowledged, falling below the B-rating during a period of approximately three years prior to default.

The number of cases of performing companies considered is about twice the number of defaulted companies: 100 cases. The sample was taken so as to include companies characterized by various levels of credit standing (approximated by the rating), industry sectors and time periods. The analysis has been substantially limited to the US market due to data availability, but we also included in the sample a few

³⁰ The actual number of months prior to default is indicated in the tables and differs from case to case, depending on when bankruptcy was declared relative to the latest closing date of accounts.

companies from other countries, just to show that the same methodology and assumptions produce similar results for companies outside the U.S..

In a certain number of cases, it was not possible to determine PD estimates for one or more models for some of the time periods, due to lack of data or to the non-comparability of balance sheet data (necessary for the RAPD and Z-Score models) between one period and another owing to mergers and acquisitions occurring in the meantime.

In applying the RAPD forecasting model ex-post, we tried to reproduce an extremely simplified implementation paradigm scheme that incorporates none of the particular forecasting capacities of a hypothetical analyst. In fact, given that at the time of analysis, the values of the basic balance sheet variables would most likely have already been known, the adoption of any particularly sophisticated forecasting scheme could be considered tainted by the ex-post awareness of said values.

Appendix A provides greater detail regarding the hypotheses adopted for development of the RAPD model in the context of comparative back-testing analysis. To put it succinctly, the RAPD implementation paradigm utilized is based on three fundamental criteria:

1. The trends of the company's economic and financial conditions are determined largely on the basis of the latest financial statement available at the time in which PD estimation is carried out. Thus, in addition to serving as the point of departure for analysis, the latest historical financial statement also determines the company's central economic and financial conditions, i.e. the trending conditions with respect to which random optimistic/pessimistic scenarios are generated (for stochastic variables, with the exception of revenues, the latest financial statement available determines the average values of the distribution functions).
2. The variability of economic and financial conditions in companies within the industry sector of reference (we use GICS classifications) is utilized to model risk for the three main stochastic variables: revenues, costs, Capex and Working Capital. Therefore, the characteristics of companies in the industry sector determine these variables' field of variation.
3. Market values observable at the moment that the analysis is presumed to have occurred, relative to real GDP growth, inflation rate, interest rates and credit spread, determine the value of a few variables, namely: average sales growth rate, cost of debt and interest rate on cash balance.

Before revealing the results, we should note once again that different assumptions would of course have provided different results; the detection of the best set of implementative assumptions requires further research and goes beyond the scope of this paper. We have worked on developing better and more evolved set of assumptions, but for the reasons outlined above, we decided to keep the RAPD implementation for the back-testing comparative analysis as simple as possible.

5.2. RAPD comparative back-testing results: defaulted companies

Table 1 below shows analytical data regarding PD at one year for each individual case at various points in time more or less distant from default, expressed in number of months; the first block is about one year before the default; the second two years before and the third three years before. Comparison of the results in the various tables shows the variation in risk estimates over time. The table's final column ("Models Median PD") shows the median PD estimate provided by all of the various models for each case, thus highlighting how much each model diverges from median values. The two bottom lines provide the median PD of the sample for each model. For the defaulted companies, interpretation of the comparative back-testing results is easier and more straightforward. In fact, in this case the model that provides the highest PD estimate earlier in time can be definitively considered as the model that has the best default predictive power.

Looking at the overall results of the sample, we draw the following conclusions:

- RAPD median PDs for the sample are by far the highest among the six models considered, and for all of the three time periods considered (1, 2 and 3 years before default). RAPD median PDs 1 year before default are above 60%, but very high average and median PDs are also registered 2 and 3 years before default.

- Z-score is the model that, after RAPD, provides the highest average and median PDs in the sample, with significant levels of PD measured over the three time periods considered.
- S&P rating implied PDs tend to be the lowest among the models considered; furthermore, a median PD 1 year before default of only 2.6% clearly shows the rating agency's failure to promptly catch the effective risk of these companies. Analysis comparing the other models' results also shows that, at various time intervals from default, the PD implicit in ratings were in nearly every case inferior to estimates derived from the other models; in almost no case do PDs implicit in ratings prove to be higher than the median values of the six models. Consider that 1 year before default more than 10% of the companies in the sample had an investment grade rating (PD lower than 0.38%); 2 years before default 24%; and 3 years before default about 50% of the companies had a PD lower than 1%.
- Moody's KMV provided PDs well below those measured by RAPD and Z-score, albeit higher than S&P ratings; the average and median PD 1 year before default are slightly above 10%, and 2 and 3 years before default they show very low levels. Similarly, DRSK reports PDs even lower than Moody's KMV (although for DRSK, we had fewer records available since the model has only recently been published). This indicates that this class of models also seems inadequate to promptly associate a high level of risk to companies destined for default; market prices in these cases did not reflect the effective risk of these companies.
- CDS implied PDs do not allow us to draw definitive significant conclusions due to lack of observations, however, the few cases reported show quite high PD values.

Looking at the individual cases in the sample, we can draw the following conclusions:

- The RAPD model provided the highest PDs in almost all the cases at 1 and 2 years before default, and in 70% of the cases at 3 years before default.
- RAPD PDs tend to increase over time consistently with the evolution of risk toward default.
- The only cases in which RAPD provided PDs lower than 10% at 1 year before default are Parmalat and Federal Mogul; however, even in these cases PD values are not negligible.
- RAPD provided very low PDs (lower than 1%) at 2 years before default in 5 cases: VeraSun Energy, Winn-Dixie, Amerco, Federal Mogul, and Gen Tek. Consider that in these cases all models measured very low PDs, with the exception of Z-score, which provided high PDs for Amerco and Federal Mogul; for the latter Moody's KMV and DRSK shows PDs around 5%.
- The Z-score model overall performance in the sample is not bad, but looking at individual companies, we noticed that, at 1 year before default, in 11 cases (about 24% of the sample) the model provided PD estimates that were too low, barely above 1%.
- It is interesting to note that in the cases of Enron, WorldCom, Parmalat and other companies in the sample, PD estimates obtained with RAPD were significantly higher than those generated by other models several years prior to default, even though said estimates were obtained on the basis of accounting data that later proved to be false. However, on closer inspection, if we analyze these companies' official reports (those that later proved false), we can easily verify that said companies were anything but flourishing, and in fact had very narrow and rapidly decreasing profit margins³¹. Therefore, if we start out with companies' conditions at the very edge of sustainability in terms of fundamentals, we can easily see how, submitting such situations to stress testing projections with RAPD, we were nevertheless able to determine very high PD estimates.

³¹ For example, considering the most remarkable case of Enron, the official year 2000 financial statement figures (those that would later have been revealed to be false) showed shrinking and very thin margins with Ebitda at 2.79% and Ebit at 1.94%, and a cash flow statement showed that over \$100 billion of revenue the gross cash flow (before capex) was less than one billion. A steady state DCF valuation of the company's EV (value of existing assets – without growth) made in the year 2000 would have been about \$ 12 billion versus a net Debt of \$ 9 billion, minority interest of \$ 2 billion and preferred stocks of \$ 2 billion. Therefore, the fundamental value of the company's equity at that time would have been inexistent, while market capitalization was \$ 60 billion one year before default and \$ 40 billion just 6 months before default, with an investment grade rating of BBB+ up to 3 months before default.

Tab. 1: PD Estimates Defaulted Companies

Company	Date of Default (m-y)	No. Months Prior to Default	1-Yr Probability of Default						Models Median PD	No. Months Prior to Default	1-Yr Probability of Default						Models Median PD	No. Months Prior to Default	1-Yr Probability of Default						Models Median PD
			RAPD	Altman Z-Score	Moody's KMV	Bloomberg DRSK	Rating S&P	CDS Implied PD			RAPD	Altman Z-Score	Moody's KMV	Bloomberg DRSK	Rating S&P	CDS Implied PD			RAPD	Altman Z-Score	Moody's KMV	Bloomberg DRSK	Rating S&P	CDS Implied PD	
Enron	Dec-01	11	24.650%	0.800%	0.250%	0.155%	0.160%	n/a	0.250%	23	26.730%	0.800%	0.020%	0.080%	0.160%	n/a	0.160%	35	13.685%	5.880%	0.090%	0.080%	0.160%	n/a	0.160%
Pilgrim's Pride	Nov-08	10	27.590%	0.230%	0.340%	0.456%	1.300%	n/a	0.456%	22	44.940%	0.230%	0.090%	n/a	0.800%	n/a	0.515%	34	0.405%	0.090%	n/a	n/a	0.800%	n/a	0.405%
Parmalat	Dec-03	11	4.020%	27.390%	0.180%	0.701%	0.380%	n/a	0.701%	23	3.885%	5.880%	0.060%	0.200%	0.38%	n/a	0.380%	35	5.695%	27.390%	n/a	0.250%	n/a	n/a	5.695%
Worldcom	Jul-02	6	61.370%	0.000%	1.700%	0.731%	0.160%	n/a	0.731%	18	2.700%	0.000%	0.680%	1.078%	0.120%	n/a	0.680%	30	0.195%	1.300%	n/a	n/a	n/a	n/a	0.748%
Delphi	Oct-05	9	61.170%	27.390%	1.650%	0.370%	0.550%	1.692%	1.671%	21	27.220%	27.390%	0.860%	0.350%	0.380%	n/a	0.860%	33	3.135%	5.880%	1.380%	0.380%	0.230%	n/a	1.380%
VeraSun Energy	Nov-08	10	75.645%	0.550%	2.120%	n/a	2.600%	n/a	2.360%	22	0.000%	0.000%	0.190%	0.320%	2.600%	n/a	0.190%	34	22.535%	0.550%	n/a	n/a	9.120%	n/a	9.120%
Key3Media	Dec-02	12	79.670%	5.880%	2.750%	2.503%	1.300%	n/a	2.750%	24	2.665%	0.090%	0.860%	n/a	1.300%	n/a	1.080%	36	n/a	n/a	n/a	n/a	n/a	n/a	n/a
IT Group	Jan-02	13	21.665%	0.800%	11.050%	3.549%	0.800%	n/a	3.549%	25	15.805%	0.550%	6.140%	1.570%	0.800%	n/a	1.565%	37	n/a	n/a	n/a	n/a	n/a	n/a	n/a
Smurfit-Stone Cont.	Feb-09	13	15.635%	27.390%	0.610%	n/a	2.600%	4.753%	4.753%	25	32.295%	27.390%	0.200%	n/a	5.880%	n/a	16.635%	37	79.685%	27.390%	n/a	n/a	5.880%	n/a	27.390%
Winn-Dixie	Feb-05	7	54.285%	1.300%	1.690%	5.450%	5.880%	n/a	5.450%	19	0.340%	0.380%	0.940%	1.080%	0.550%	n/a	0.550%	31	0.340%	0.380%	n/a	0.240%	0.380%	n/a	0.188%
Tousa	Jan-08	12	15.600%	0.800%	n/a	n/a	5.880%	n/a	5.880%	24	5.235%	0.230%	n/a	n/a	2.600%	n/a	2.600%	36	9.660%	0.230%	n/a	n/a	2.600%	n/a	2.600%
Amerco	Oct-02	7	77.395%	9.120%	0.440%	2.007%	5.880%	n/a	5.880%	19	0.135%	100.000%	1.000%	1.020%	5.880%	n/a	1.015%	31	0.001%	5.880%	1.970%	0.890%	1.300%	n/a	1.300%
Polaroid	Jul-01	7	26.545%	1.300%	11.080%	5.999%	1.300%	n/a	5.999%	19	25.380%	1.300%	2.350%	0.660%	0.800%	n/a	1.300%	31	77.650%	5.880%	3.120%	1.870%	0.800%	n/a	3.120%
Exide Technologies	Apr-02	12	67.715%	100.000%	7.190%	1.093%	2.600%	n/a	7.190%	24	82.165%	27.390%	13.140%	2.510%	1.300%	n/a	13.140%	36	80.770%	27.390%	8.420%	3.120%	1.300%	n/a	8.420%
Northwest Airlines	Sep-05	8	98.515%	52.335%	7.470%	5.246%	5.880%	n/a	7.470%	20	96.695%	37.861%	6.120%	1.170%	2.600%	n/a	6.120%	32	98.555%	52.335%	13.060%	4.180%	1.300%	n/a	13.060%
Swissair	Oct-01	9	34.675%	15.805%	0.050%	n/a	0.120%	n/a	7.963%	21	17.835%	1.300%	0.340%	n/a	0.120%	n/a	0.820%	33	18.415%	2.600%	n/a	n/a	n/a	n/a	10.508%
Dana	Mar-06	2	81.495%	27.390%	10.200%	12.108%	2.600%	n/a	12.108%	14	33.790%	27.390%	0.620%	0.440%	0.380%	n/a	0.620%	26	47.940%	27.390%	0.310%	0.480%	0.800%	n/a	0.800%
Dura	Oct-06	9	28.620%	0.800%	20.000%	n/a	5.880%	n/a	12.940%	21	10.580%	0.800%	8.440%	n/a	2.600%	n/a	5.520%	33	12.480%	0.800%	4.080%	n/a	1.300%	n/a	2.690%
Kevco	Feb-01	13	63.835%	0.020%	20.000%	n/a	5.880%	n/a	12.940%	25	46.850%	0.020%	6.030%	n/a	2.600%	n/a	4.315%	37	n/a	n/a	n/a	n/a	n/a	n/a	n/a
Newcor	Sep-01	8	55.420%	0.800%	20.000%	n/a	5.880%	n/a	12.940%	20	56.280%	0.800%	20.000%	n/a	5.880%	n/a	12.940%	32	55.855%	0.800%	11.450%	n/a	2.600%	n/a	7.025%
Healthsouth	Apr-03	3	19.640%	72.343%	7.560%	n/a	0.800%	n/a	13.600%	15	59.770%	52.335%	1.520%	n/a	0.380%	n/a	26.928%	27	n/a	n/a	n/a	n/a	n/a	n/a	n/a
AMR	Nov-11	10	58.790%	52.335%	19.170%	1.257%	9.120%	8.674%	14.145%	22	99.540%	52.335%	24.810%	1.380%	9.120%	16.568%	20.689%	34	95.735%	52.335%	27.040%	10.500%	9.120%	28.830%	27.935%
Calpine	Dec-05	11	100.000%	15.805%	17.810%	3.488%	5.880%	n/a	15.805%	23	79.860%	9.120%	20.000%	1.060%	5.880%	n/a	9.120%	35	76.275%	15.805%	20.000%	2.770%	0.800%	n/a	15.805%
Tronox	Mar-08	2	84.490%	27.390%	4.420%	n/a	2.600%	n/a	15.905%	14	36.630%	27.390%	0.800%	n/a	1.300%	n/a	14.345%	26	5.535%	27.390%	n/a	n/a	1.300%	n/a	5.535%
Federal Mogul	Oct-01	8	8.645%	27.390%	20.000%	16.176%	2.600%	n/a	16.176%	20	0.110%	27.390%	4.370%	5.560%	0.550%	n/a	4.370%	32	1.885%	27.390%	0.470%	0.980%	0.550%	n/a	0.980%
Mirant	Jul-03	6	23.060%	15.805%	20.000%	n/a	0.800%	n/a	17.903%	18	0.070%	15.805%	13.710%	4.640%	0.380%	n/a	4.640%	30	n/a	n/a	n/a	n/a	n/a	n/a	n/a
Covanta Energy	Mar-02	2	61.850%	27.390%	20.000%	8.250%	5.880%	n/a	20.000%	14	99.315%	15.805%	2.210%	0.240%	0.160%	n/a	2.210%	26	100.00%	5.880%	3.520%	4.930%	0.160%	n/a	4.930%
Solutia	Dec-03	12	66.240%	27.390%	14.270%	n/a	0.800%	n/a	20.830%	24	98.605%	27.390%	n/a	0.890%	0.230%	n/a	14.139%	36	75.490%	27.390%	n/a	1.320%	0.230%	n/a	14.355%
Gen Tek	Oct-02	10	91.480%	27.390%	20.000%	n/a	1.300%	n/a	23.695%	22	0.780%	0.800%	2.130%	n/a	0.800%	n/a	0.800%	34	7.325%	5.880%	1.360%	n/a	0.800%	n/a	3.620%
Tribune	Dec-08	11	28.540%	72.343%	0.210%	n/a	5.880%	26.976%	26.976%	23	4.995%	37.861%	0.050%	0.080%	0.550%	n/a	0.550%	35	1.320%	0.550%	n/a	0.040%	0.120%	n/a	0.335%
Sea Containers	Oct-06	22	99.985%	27.390%	n/a	n/a	1.300%	n/a	27.390%	34	30.96%	27.39%	n/a	n/a	1.30%	n/a	27.390%	46	n/a	n/a	n/a	n/a	n/a	n/a	n/a
AbitibiBowater	Apr-09	3	99.880%	27.390%	35.000%	20.030%	9.120%	n/a	27.390%	15	100.000%	100.000%	35.000%	6.620%	n/a	n/a	67.500%	27	n/a	n/a	n/a	n/a	n/a	n/a	n/a
Citadel Broadcasting	Dec-09	11	28.360%	100.000%	35.000%	6.770%	2.600%	n/a	28.360%	23	28.590%	72.343%	12.490%	14.930%	2.600%	n/a	14.930%	35	0.815%	15.805%	0.340%	n/a	n/a	n/a	0.815%
United Airlines	Dec-02	11	100.000%	52.335%	7.800%	n/a	2.600%	n/a	30.068%	23	35.725%	15.805%	1.600%	n/a	0.550%	n/a	8.703%	35	0.001%	5.880%	0.830%	n/a	0.550%	n/a	0.690%
Six Flags Inc.	Apr-09	3	46.495%	100.000%	n/a	8.260%	15.805%	n/a	31.150%	15	92.825%	100.000%	n/a	8.590%	15.805%	n/a	54.315%	27	89.655%	100.000%	n/a	n/a	9.120%	n/a	89.655%
Spectrum Brands	Apr-09	6	40.340%	27.390%	35.000%	n/a	15.805%	n/a	31.195%	18	25.805%	5.880%	32.100%	n/a	15.805%	n/a	20.805%	30	23.265%	0.800%	n/a	n/a	9.120%	n/a	9.120%
Chemtura	Mar-09	2	43.575%	100.000%	25.270%	11.320%	5.880%	63.285%	34.423%	14	20.735%	27.390%	0.560%	0.720%	0.550%	3.764%	2.241%	26	7.870%	27.390%	n/a	0.180%	0.550%	n/a	4.210%
Visteon	May-09	4	81.425%	100.000%	35.000%	9.620%	9.120%	n/a	35.000%	16	49.050%	27.390%	8.250%	6.560%	5.880%	19.462%	13.856%	28	49.265%	27.390%	1.870%	0.590%	5.880%	3.712%	4.796%
Qwest Communic.	Dec-02	12	99.510%	37.861%	n/a	n/a	0.160%	n/a	37.861%	24	1.775%	5.880%	n/a	n/a	0.160%	n/a	1.775%	36	n/a	n/a	n/a	n/a	n/a	n/a	n/a
Energis	Aug-02	17	99.075%	72.343%	0.490%	n/a	9.120%	n/a	40.732%	29	99.555%	52.335%	0.020%	n/a	9.120%	n/a	30.728%	41	n/a	n/a	n/a	n/a	n/a	n/a	n/a
Air Canada	Apr-03	3	94.090%	72.343%	17.190%	n/a	2.600%	n/a	44.767%	15	99.870%	27.390%	20.000%	6.180%	2.600%	n/a	20.000%	27	n/a	n/a	n/a	3.590%	n/a	n/a	3.590%
IFCO Systems	Mar-02	2	100.000%	100.000%	7.760%	n/a	5.880%	n/a	53.880%	14	99.615%	27.390%	2.280%	10.380%	1.300%	n/a	10.375%	26	n/a	n/a	n/a	n/a	n/a	n/a	n/a
Newpage	Jul-11	6	99.890%	n/a	n/a	n/a	15.805%	n/a	57.848%	18	100.000%	n/a	n/a	n/a	15.805%	n/a	57.903%	30	68.14%	n/a	n/a	n/a	5.880%	n/a	37.008%
Mississippi Chemical	May-03	11	99.540%	100.000%	20.000%	n/a	5.																		

Data source: Bloomberg, Moody's KMV, Standard & Poor's. – RAPD Processing: by wiredex

To underscore the differences among the results provided by the various measures considered, Figures 2-4 show, for three of the most significant and best-known cases in the sample, the evolution of PD at one year, estimated using the various models beginning from approximately two years prior to the declaration of bankruptcy.

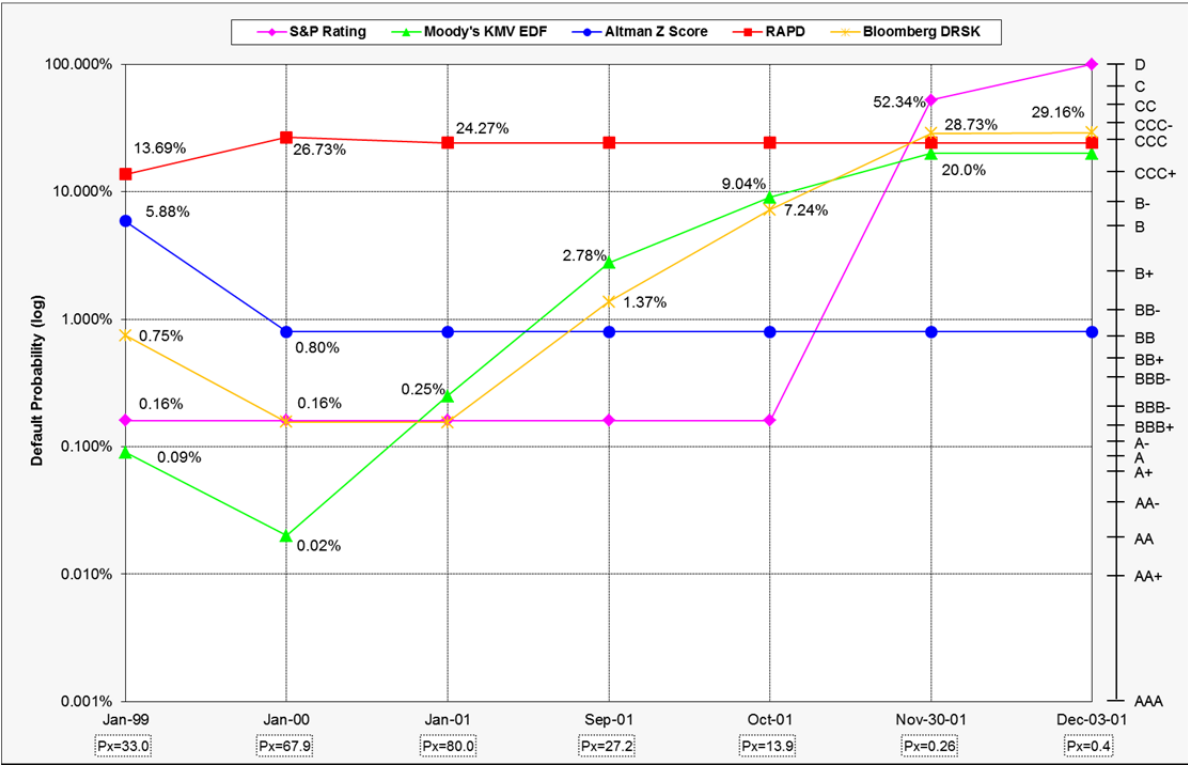
We decided to show, for each year, PD values estimated at the start of said year and at every successive interval at which variations in the companies' ratings occurred. Since PDs estimated with the RAPD and Z-Score models only vary depending on the availability of balance sheet data, estimates generated with these models remained constant throughout the year of reference; hence the absence of revision of estimates over the course of the year. PDs associated with ratings, and the Moody's KMV model, on the other hand, are values estimated on the specific date indicated, thus their evolution also highlights possible revisions that occurring over the course of the year.

PD is reported for each period in logarithmic scale on the y-axis on the left, while the corresponding rating class is indicated on the right. The rating *master scale* is shown purely to indicate the association between a PD and a corresponding rating class, and is based on historical frequencies of default reported in charts published by S&P.

The limitations of graphic representation make it impossible to show a constant time scale, hence the distance between the various points does not adequately emphasize the temporal dynamics; to better appreciate them, we must note the dates of reference indicated on the horizontal axis. Otherwise, we may fail to immediately notice that, in many of the cases presented, revisions of companies' levels of risk (in particular Moody's KMV, Bloomberg DRSK and S&P) occurred only when bankruptcy was already imminent. Beneath each date, each company's corresponding stock price are shown, so as to give an idea of the parallel evolution of its market value; thus allowing us to better comprehend the results of the Moody's KMV model and Bloomberg DRSK, which are highly dependent on market price values.

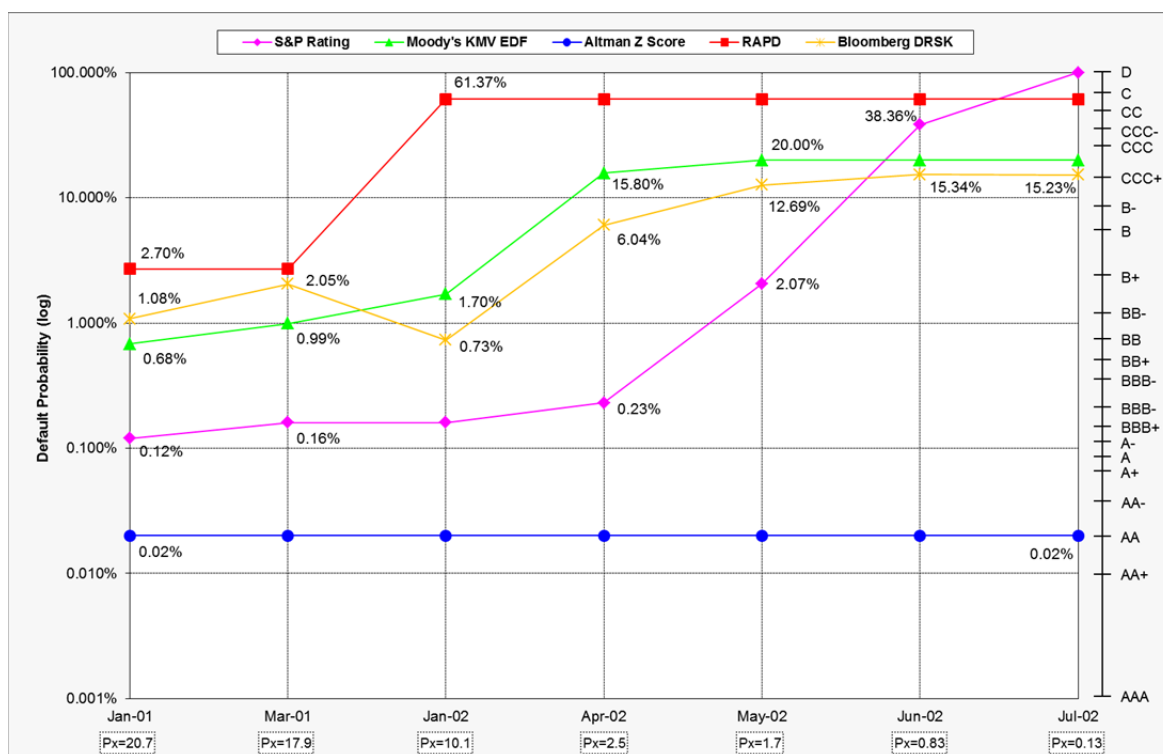
Examination of the evolution of PDs in the individual cases clearly shows that rating agencies and the market often significantly readjusted their judgments on companies' level of risk quite brusquely only a few months prior to default.

Fig. 2 - Credit risk models comparative test: Enron



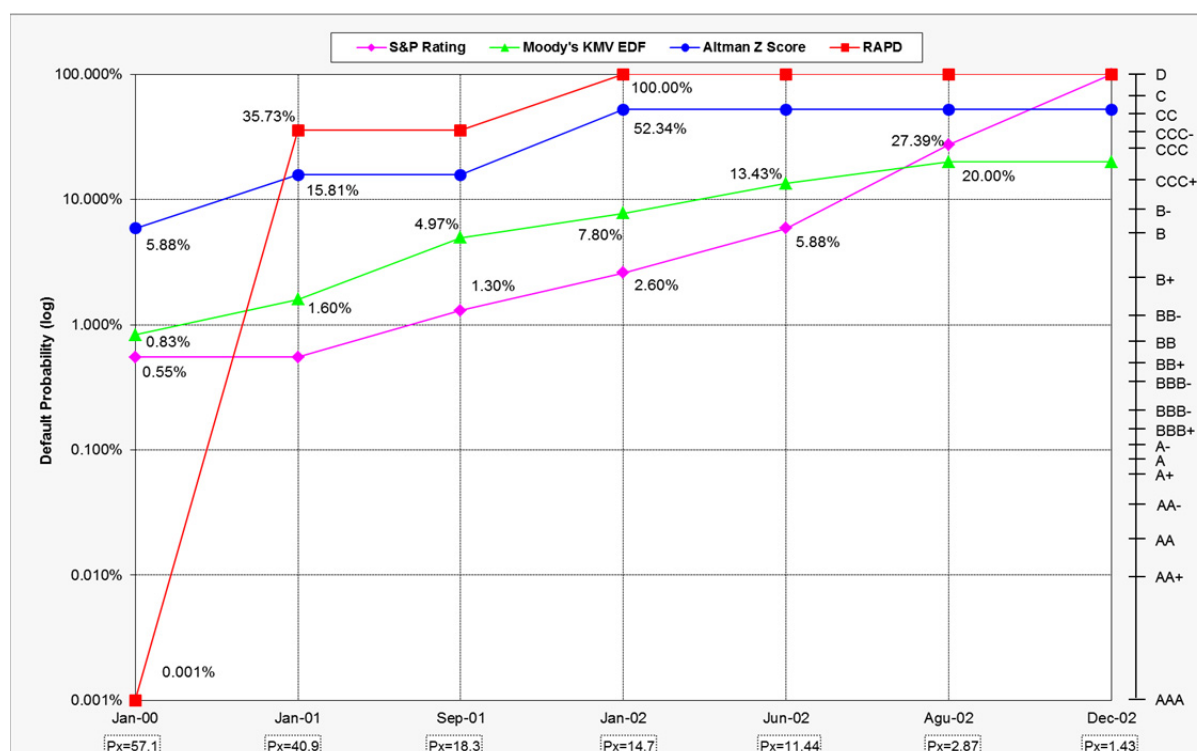
Data source: Bloomberg, Moody's KMV, Standard & Poor's. - RAPD Processing: by wiredex and value corporate [Bloomberg APPS VCORP<GO>].

Fig. 3 - Credit risk models comparative test: WorldCom



Data source: Bloomberg, Moody's KMV, Standard & Poor's. - RAPD Processing: by wiredex/value.corporate [Bloomberg APPS VCORP<GO>].

Fig. 4 - Credit risk models comparative test: United Airlines



Data source: Bloomberg, Moody's KMV, Standard & Poor's. - RAPD Processing: by wiredex/value.corporate [Bloomberg APPS VCORP<GO>].

5.3. RAPD comparative back-testing results: performing companies

The performing companies sample has overall about twice as many cases as the defaulted companies sample; the composition of the sample was driven by the need to make it as heterogeneous as possible, considering companies with different credit standings (represented by the ratings) and belonging to different industry sectors; we also considered different time periods (divided into three periods of three years each) in order to cover different economic conditions occurring between 2003 and 2011.

The results of the analysis are shown in Tables 2 (2003-2005), 3 (2006-2008), and 4 (2009-2011) below, according to the three time periods considered. Records are reported in the same way as described for the defaulted companies; Moody's KMV data were not available for the 2003-2005 period. Furthermore, to highlight the "official" credit standing of the companies, implied PDs assigned to non-investment-grade ratings (below BBB-) are underscored in the "Rating S&P" column.

The performances of the different credit risk models are more difficult to assess, and the interpretation of the results is unavoidably less univocal for performing companies than for defaulted companies. However, we believe that this analysis permits at least two types of important tests:

1. Verifying whether a model is systematically up/downward biased; in the case of the RAPD model (which provided the highest PDs for defaulted companies), we want to determine whether it is upward biased, i.e. whether it persistently provides very high PD values.
2. Checking for potentially incorrect PD estimates by looking at each individual case to see if there is any apparently anomalous PD estimate; for example a PD that seems particularly high for a company that did not default may indicate an incorrect risk estimate.

The back-testing results show that the RAPD model is not upward biased; in fact the RAPD median PD values for all the time periods considered are not higher than those registered by the other models, and looking at individual cases as well, we see that RAPD estimates are in line with the median values of all the models considered ("Models Median PD" column).

Checking individual companies, we detect just a few potential anomalous cases of very high PD values (above 10%) in the RAPD model which call for comment.

In Table 2 (2003-2005), Eastman Kodak shows RAPD PD estimates in 2004 and 2005 above 4%, a value that is not extremely high but is quite a bit higher than that recorded by the other models, and Unisys shows a 2005 RAPD PD of 38%, with very high PDs also reported by Z-score for this company since 2003. Both companies were operating in those years with very low and progressively shrinking Ebitda margins; years later, Eastman Kodak defaulted (in 2012) and S&P put Unisys in "Selective Default" (CC rating with negative outlook) in 2009. So for these two companies, the two models did not seem to have provided excessively high PDs compared to their actual level of risk; it could instead be argued that they correctly measured a risk of future credit deterioration, which in fact occurred. Alcatel shows a very high RAPD PD in 2003, which diminishes in the two following years; nevertheless, this dynamic follows the evolution of the company's economic and financial conditions, which shifted from negative Ebitda margins in 2003 and 2004 to a positive net income in 2005. Tenneco shows very high PDs with RAPD and Z-score during the entire period analyzed, notwithstanding the fact that the company had a low rating (B in 2003-2004 and B+ / BB- in 2005); in this case, it seems that the two models provided PDs that were apparently too high and therefore incorrect.

In Table 3 (2006-2008) there are five companies that showed high PDs with the RAPD model for at least one year: Shiloh Industries, Pantry, AES, Vail Resort and Quicksilver. These companies all had ratings below investment grade, and in two cases (Vail Resort and AES) Z-score also reported high PDs. These cases are all characterized by companies operating with very low Ebitda margins; considering the simple RAPD implementation hypothesis applied in back-testing, according to which the current economic and financial conditions of the companies are stressed through Monte Carlo simulation, PDs estimated with RAPD are strongly affected by the figures recorded in the last financial statement (which represents the central scenario of the simulation), such as a low or negative Ebitda margin. In these circumstances, many of the scenarios generated in the simulation easily lead to the default event, and this explains the high PD estimates.

In Table 4 (2009-2011) we see the following eight cases of potentially incorrect RAPD estimates in at least one year of the time period considered: Con-way, MeadWestvaco, Micron Technology Inc., Advanced Micro Devices, Consolidated Communications, Alaska Comm. Systems Group and Valhi. For

most of these companies, Z-score also provided very high PD measures, and in some cases S&P (Advanced Micro Devices and Valhi) and CDS (Advanced Micro Devices) did as well. For these cases the same consideration made for the anomalous cases of Table 3 stands.

Looking at the results provided by *a la Merton* models (Moody's KMV and Bloomberg DRSK), we noticed that they provide, with few exceptions, PD estimates that are in almost all cases very low and quite similar, denoting a very low capacity for risk discrimination among the companies in the sample.

6. Conclusions

The RAPD approach has the capacity to estimate a company's expected PD in a way that is methodologically consistent with the theory of finance, considering an objective causal link between the technical condition of financial insolvency and the status of default. It is a closed model, in the sense that the two variables on which the condition of default depends – stock of debt and EV – are calculated endogenously by the model itself, implicitly incorporating the indisputable economic rule according to which debt at the end of each period is a function first and foremost of the company's financial needs, while a company's debt capacity varies according to the present value of its expected cash flows.

The method can be applied to any type of company, public or private companies. PD estimates obtained tie in with the time horizon relevant to the lender, and are thus particularly suited to measuring mid-to-long-term financial risk. In particular, the method can also be used to estimate PD in project financing operations³².

The RAPD approach also allows users to obtain a direct estimate of *loss given default* (LGD) and, through reconstruction of its probability distribution, *expected loss* (EL) and *unexpected loss* (UL). Opportunely extending the model, it will also be possible to manage situations characterized by guarantees and different degrees of creditor seniority.

Preliminary empirical evidence has proven encouraging in supporting the validity of the RAPD approach, even when applied with simplified and standardized implementation paradigms. Comparative back-testing analyses show that the RAPD model performs better than other widespread credit risk measures in estimating the risk of default, providing relevant early warning signals years before the event of default; results also prove that the RAPD model is not upward-biased in estimating PD.

In conclusion, we would like to highlight that, in addition to the particular standards of implementation we chose to apply in this study (affected by the scope of the exercise), the RAPD approach proposed should be considered a model that, in our opinion, allows an analyst to correctly approach a firm's credit risk analysis with a sound methodology in accordance with the theory of finance, providing the possibility to exploit all the best available information at hand in specific cases. Of course, the actual results of practical application of the RAPD model may be correct or incorrect depending on the assumptions and modalities of implementation adopted, to the same extent that the result of a company valuation made by a DCF model – while perfectly correct in its theoretical background – may be adequate or inadequate depending on the assumptions made.

³² We believe that the simulation approach is the only satisfactory solution that conforms to the tenets of the Basel Committee document on specialized lending; see Basel Committee on Banking Supervision (2001), p. 16.

Tab. 2: PD Estimates Performing Companies 2003/2005

Company	Probability of Default 2005						Models Median PD
	RAPD	Altman Z-Score	Moody's KMV	Bloomberg DRSK	Rating S&P	CDS Implied PD	
Avon Products	0.001%	0.000%	n/a	0.012%	0.090%	n/a	0.006%
Estee Lauder Companies	0.020%	0.000%	n/a	0.010%	0.070%	n/a	0.015%
Coca Cola	0.001%	0.040%	n/a	0.002%	0.070%	n/a	0.021%
Whole Foods Market	0.001%	0.010%	n/a	0.034%	<u>0.550%</u>	n/a	0.022%
Du Pont (EI) De Nemours	0.020%	0.230%	n/a	0.016%	0.040%	n/a	0.030%
Ralph Lauren	0.001%	0.020%	n/a	0.050%	0.230%	n/a	0.035%
Centurylink	0.025%	0.040%	n/a	0.044%	0.160%	0.333%	0.044%
FedEx	0.060%	0.040%	n/a	0.035%	0.230%	n/a	0.050%
Enel SpA	0.001%	5.880%	n/a	0.004%	0.070%	0.124%	0.070%
Pss World Medical	0.001%	0.020%	n/a	0.132%	<u>1.300%</u>	n/a	0.076%
EDF	0.001%	15.800%	n/a	n/a	0.040%	0.120%	0.080%
Honeywell International	0.645%	0.230%	n/a	0.019%	0.090%	0.123%	0.123%
Masco	1.395%	0.090%	n/a	0.024%	0.160%	n/a	0.125%
Omnicom Group	0.590%	1.300%	n/a	0.012%	0.120%	0.143%	0.143%
Jones Group	0.850%	0.000%	n/a	0.067%	0.230%	n/a	0.148%
Supervalu	0.975%	0.040%	n/a	0.069%	0.230%	n/a	0.149%
Continental	0.001%	0.230%	n/a	0.016%	0.160%	0.172%	0.160%
Telecom Italia SpA	0.335%	0.380%	n/a	0.009%	0.160%	0.200%	0.200%
Stoneridge	0.001%	0.230%	n/a	0.209%	1.300%	n/a	0.219%
Mc Kesson	0.500%	0.230%	n/a	0.034%	0.230%	n/a	0.230%
Alcatel Lucent	0.001%	27.390%	n/a	0.436%	<u>0.800%</u>	0.454%	0.454%
Frontier Communications	0.545%	5.880%	n/a	0.105%	<u>0.550%</u>	0.320%	0.545%
Eastman Kodak	4.535%	1.300%	n/a	n/a	0.380%	0.692%	0.996%
Iron Mountain	1.240%	1.300%	n/a	0.107%	<u>1.300%</u>	n/a	1.270%
Cincinnati Bell	0.200%	100.000%	n/a	0.577%	<u>2.600%</u>	n/a	1.588%
Unisys	38.090%	27.390%	n/a	0.459%	<u>0.550%</u>	n/a	13.970%
Tenneco	30.165%	27.390%	n/a	0.489%	<u>2.600%</u>	n/a	14.995%
Median	0.060%	0.230%	n/a	0.035%	0.230%	0.186%	0.143%

Company	Probability of Default 2004						Models Median PD
	RAPD	Altman Z-Score	Moody's KMV	Bloomberg DRSK	Rating S&P	CDS Implied PD	
	0.005%	0.020%	n/a	0.012%	0.090%	n/a	0.016%
	0.045%	0.020%	n/a	0.034%	0.070%	n/a	0.040%
	0.001%	0.040%	n/a	0.001%	0.070%	n/a	0.021%
	0.001%	0.010%	n/a	0.034%	<u>0.550%</u>	n/a	0.022%
	0.870%	5.880%	n/a	0.016%	0.040%	n/a	0.455%
	0.005%	0.020%	n/a	0.160%	0.230%	n/a	0.090%
	0.045%	0.040%	n/a	0.074%	0.160%	n/a	0.060%
	0.070%	0.160%	n/a	0.131%	0.230%	n/a	0.146%
	0.001%	5.880%	n/a	0.012%	0.070%	0.389%	0.070%
	3.550%	0.040%	n/a	0.765%	<u>1.300%</u>	n/a	1.033%
	0.001%	9.120%	n/a	n/a	0.040%	0.269%	0.155%
	0.010%	0.230%	n/a	0.027%	0.090%	n/a	0.058%
	0.605%	0.090%	n/a	0.038%	0.160%	n/a	0.125%
	0.695%	1.300%	n/a	0.040%	0.120%	n/a	0.408%
	0.030%	0.000%	n/a	0.067%	0.230%	n/a	0.048%
	0.720%	0.160%	n/a	0.380%	0.230%	n/a	0.305%
	0.001%	0.800%	n/a	0.016%	0.230%	0.349%	0.230%
	0.385%	0.550%	n/a	0.009%	0.160%	0.791%	0.385%
	0.005%	0.090%	n/a	0.234%	<u>1.300%</u>	n/a	0.162%
	4.525%	0.120%	n/a	0.107%	0.230%	n/a	0.175%
	9.320%	27.390%	n/a	0.158%	<u>2.600%</u>	1.920%	2.600%
	3.435%	5.880%	n/a	0.193%	0.230%	n/a	1.833%
	4.425%	5.880%	n/a	n/a	0.380%	1.152%	2.788%
	0.635%	5.880%	n/a	0.077%	<u>1.300%</u>	n/a	0.968%
	0.605%	100.00%	n/a	0.296%	<u>2.600%</u>	n/a	1.603%
	0.095%	27.390%	n/a	0.123%	<u>0.550%</u>	n/a	0.336%
	30.53%	27.390%	n/a	0.216%	<u>5.880%</u>	n/a	16.635%
Median	0.095%	0.230%	n/a	0.074%	0.230%	0.590%	0.175%

Company	Probability of Default 2003						Models Median PD
	RAPD	Altman Z-Score	Moody's KMV	Bloomberg DRSK	Rating S&P	CDS Implied PD	
	0.085%	0.090%	n/a	0.019%	0.090%	n/a	0.088%
	0.045%	0.000%	n/a	0.057%	0.070%	n/a	0.051%
	0.001%	0.040%	n/a	0.010%	0.070%	n/a	0.025%
	0.001%	0.040%	n/a	0.055%	<u>0.550%</u>	n/a	0.047%
	0.000%	0.800%	n/a	0.088%	0.040%	n/a	0.064%
	0.055%	0.090%	n/a	0.100%	0.230%	n/a	0.095%
	0.130%	0.120%	n/a	0.138%	0.160%	n/a	0.134%
	0.070%	0.120%	n/a	0.208%	0.230%	n/a	0.164%
	0.470%	9.120%	n/a	0.143%	0.070%	n/a	0.307%
	1.060%	0.550%	n/a	0.185%	<u>1.300%</u>	n/a	0.805%
	0.050%	n/a	n/a	n/a	0.020%	n/a	0.035%
	0.010%	0.800%	n/a	0.244%	0.090%	n/a	0.167%
	0.315%	0.230%	n/a	0.214%	0.160%	n/a	0.222%
	3.155%	9.120%	n/a	0.225%	0.090%	n/a	1.690%
	0.001%	0.000%	n/a	0.139%	0.230%	n/a	0.070%
	2.330%	0.230%	n/a	0.069%	0.230%	n/a	0.230%
	0.010%	0.800%	n/a	0.088%	0.230%	n/a	0.064%
	0.205%	2.600%	n/a	0.622%	0.160%	n/a	0.414%
	0.001%	0.090%	n/a	0.821%	<u>1.300%</u>	n/a	0.456%
	6.765%	0.120%	n/a	0.041%	0.230%	n/a	0.175%
	68.630%	100.000%	n/a	0.607%	<u>2.600%</u>	n/a	35.625%
	5.035%	9.120%	n/a	0.443%	0.230%	n/a	2.739%
	0.310%	9.120%	n/a	n/a	0.160%	n/a	0.310%
	0.795%	5.880%	n/a	0.157%	<u>1.300%</u>	n/a	1.048%
	23.790%	100.000%	n/a	3.292%	<u>9.120%</u>	n/a	16.455%
	0.375%	52.340%	n/a	0.586%	<u>0.550%</u>	n/a	0.568%
	33.595%	100.000%	n/a	2.844%	<u>5.880%</u>	n/a	19.738%
Median	0.205%	0.675%	n/a	0.157%	0.230%	n/a	0.222%

Data source: Bloomberg, Moody's KMV, Standard & Poor's. – RAPD Processing: by wiredex and value.corporate [Bloomberg APPS VCORP<GO>].

Tab. 3: PD Estimates Performing Companies 2006-2008

Company	Probability of Default 2008						Models Median PD
	RAPD	Altman Z-Score	Moody's KMV	Bloomberg DRSK	Rating S&P	CDS Implied PD	
Exxon Mobil	0.001%	0.000%	0.010%	0.002%	0.000%	n/a	0.001%
Texas Instruments	0.001%	0.000%	0.010%	0.002%	0.090%	n/a	0.002%
Intel	0.001%	0.000%	0.010%	0.007%	0.070%	6.045%	0.008%
Nestlé	0.001%	0.020%	0.010%	0.000%	0.020%	n/a	0.010%
Abbott Laboratories	0.001%	0.020%	0.010%	0.005%	0.020%	0.284%	0.010%
Procter & Gamble	0.001%	0.090%	0.010%	0.001%	0.040%	n/a	0.010%
British American Tobacco	0.001%	0.090%	0.010%	0.000%	0.160%	n/a	0.010%
Altria	0.001%	0.020%	0.010%	0.001%	0.160%	0.656%	0.015%
Amphenol	0.001%	0.000%	0.020%	0.020%	0.380%	n/a	0.020%
PVH	0.010%	0.020%	0.020%	0.061%	<u>0.550%</u>	n/a	0.020%
TDK	0.500%	0.020%	0.020%	n/a	0.040%	n/a	0.020%
Merck & Co.	0.001%	0.090%	0.010%	0.007%	0.040%	0.326%	0.025%
Ibm	0.015%	1.300%	0.020%	0.020%	0.070%	0.269%	0.045%
Dell	0.001%	0.090%	0.050%	0.004%	0.090%	n/a	0.050%
Chevron	0.185%	0.090%	0.010%	0.013%	0.020%	0.311%	0.055%
Xilinx	0.001%	0.090%	0.060%	0.015%	0.380%	n/a	0.060%
Lockheed Martin	0.145%	0.020%	0.010%	0.019%	0.120%	0.219%	0.070%
Corning	0.001%	0.090%	0.060%	0.013%	0.230%	0.304%	0.075%
France Telecom	0.050%	9.120%	0.030%	0.036%	0.120%	0.492%	0.085%
Intl Rectifier	0.001%	0.090%	0.170%	0.059%	<u>0.800%</u>	n/a	0.090%
Gannett	0.695%	0.000%	0.040%	0.156%	0.120%	n/a	0.120%
STM Microelectronics	0.370%	0.230%	0.060%	0.000%	0.120%	n/a	0.120%
Exelon	0.001%	5.880%	0.040%	0.099%	0.160%	0.790%	0.129%
Valero Energy	0.050%	0.090%	0.180%	0.062%	0.230%	0.744%	0.135%
Universal Health Services	0.245%	0.020%	0.060%	0.017%	0.380%	0.486%	0.153%
Sealed Air	0.005%	0.230%	0.070%	0.097%	0.230%	0.393%	0.164%
Oxford Industries	1.275%	0.020%	0.080%	0.165%	<u>1.300%</u>	n/a	0.165%
NVR	0.200%	0.020%	0.310%	0.123%	0.380%	n/a	0.200%
Boeing	0.355%	5.880%	0.050%	0.028%	0.070%	0.474%	0.213%
Ryder System	0.285%	1.300%	0.080%	0.152%	0.230%	n/a	0.230%
Sempra Energy	1.490%	27.390%	0.050%	0.009%	0.160%	0.332%	0.246%
Safeway	0.070%	1.300%	0.040%	0.132%	0.380%	0.486%	0.256%
Steel Dynamics	4.370%	0.230%	0.340%	0.132%	<u>0.550%</u>	n/a	0.340%
Westlake Chemicals	4.480%	0.230%	0.300%	0.401%	<u>0.550%</u>	n/a	0.401%
Toshiba	0.480%	5.880%	0.120%	n/a	0.380%	n/a	0.480%
Quicksilver	2.720%	0.800%	0.280%	0.413%	<u>1.300%</u>	n/a	0.800%
Shiloh Industries	6.030%	0.800%	1.210%	0.516%	<u>1.300%</u>	n/a	1.210%
Pantry	29.85%	2.600%	1.260%	1.090%	<u>1.300%</u>	n/a	1.300%
Vail Resort	2.875%	5.880%	0.040%	0.037%	<u>1.300%</u>	n/a	1.300%
AES	18.095%	5.880%	0.100%	0.354%	<u>1.300%</u>	n/a	1.300%
Median	0.060%	0.090%	0.045%	0.024%	0.195%	0.434%	0.105%

Company	Probability of Default 2007						Models Median PD
	RAPD	Altman Z-Score	Moody's KMV	Bloomberg DRSK	Rating S&P	CDS Implied PD	
Exxon Mobil	0.001%	0.000%	0.010%	0.006%	0.000%	n/a	0.001%
Texas Instruments	0.001%	0.000%	0.010%	0.001%	0.090%	n/a	0.001%
Intel	0.001%	0.000%	0.010%	0.002%	0.070%	n/a	0.002%
Nestlé	0.001%	0.020%	0.010%	0.000%	0.000%	0.042%	0.006%
Abbott Laboratories	0.001%	0.020%	0.010%	0.002%	0.020%	n/a	0.015%
Procter & Gamble	0.001%	0.090%	n/a	0.001%	0.040%	n/a	0.021%
British American Tobacco	0.001%	0.090%	0.010%	0.000%	0.160%	0.083%	0.046%
Altria	0.001%	0.230%	0.010%	0.000%	0.160%	0.116%	0.063%
Amphenol	0.175%	0.000%	0.020%	0.019%	0.380%	n/a	0.020%
PVH	0.075%	0.020%	n/a	0.121%	<u>0.550%</u>	n/a	0.098%
TDK	0.030%	0.020%	n/a	n/a	0.040%	n/a	0.030%
Merck & Co.	0.001%	0.090%	0.010%	0.001%	0.040%	n/a	0.010%
Ibm	0.001%	1.300%	0.010%	0.002%	0.070%	0.041%	0.025%
Dell	0.001%	0.090%	n/a	0.001%	0.090%	n/a	0.046%
Chevron	0.030%	0.090%	0.010%	0.006%	0.020%	n/a	0.020%
Xilinx	0.001%	0.000%	0.060%	0.001%	0.380%	n/a	0.001%
Lockheed Martin	0.305%	0.090%	0.010%	0.004%	0.160%	n/a	0.090%
Corning	0.001%	0.090%	0.050%	0.050%	0.230%	n/a	0.050%
France Telecom	0.120%	15.800%	0.070%	0.008%	0.120%	0.081%	0.100%
Intl Rectifier	0.575%	0.230%	n/a	0.119%	<u>0.800%</u>	n/a	0.403%
Gannett	0.275%	0.000%	0.010%	0.112%	0.120%	n/a	0.112%
STM Microelectronics	0.560%	0.090%	0.030%	0.000%	0.120%	0.108%	0.099%
Exelon	0.001%	5.880%	0.010%	0.007%	0.160%	n/a	0.010%
Valero Energy	0.835%	0.020%	0.080%	0.057%	0.230%	0.134%	0.107%
Universal Health Services	0.160%	0.020%	0.040%	0.008%	0.230%	n/a	0.040%
Sealed Air	0.001%	0.230%	0.030%	0.013%	0.230%	0.142%	0.086%
Oxford Industries	1.495%	0.020%	n/a	0.143%	<u>1.300%</u>	n/a	0.722%
NVR	0.035%	0.020%	0.130%	0.049%	0.380%	n/a	0.049%
Boeing	1.900%	5.880%	0.010%	0.028%	0.070%	n/a	0.070%
Ryder System	0.555%	2.600%	0.060%	0.141%	0.230%	n/a	0.230%
Sempra Energy	0.385%	27.390%	0.030%	0.030%	0.160%	0.074%	0.117%
Safeway	0.220%	2.600%	0.020%	0.024%	0.380%	0.109%	0.164%
Steel Dynamics	0.115%	0.090%	0.110%	0.073%	<u>0.550%</u>	n/a	0.110%
Westlake Chemicals	0.001%	0.090%	0.120%	0.064%	<u>0.550%</u>	n/a	0.090%
Toshiba	0.335%	5.880%	n/a	n/a	0.380%	n/a	0.380%
Quicksilver	10.93%	0.230%	n/a	0.325%	<u>0.800%</u>	n/a	0.562%
Shiloh Industries	12.000%	0.230%	n/a	0.231%	<u>1.300%</u>	n/a	0.765%
Pantry	0.905%	0.380%	n/a	0.231%	<u>1.300%</u>	n/a	0.643%
Vail Resort	4.665%	27.390%	n/a	0.032%	<u>1.300%</u>	n/a	2.983%
AES	3.905%	9.120%	0.080%	0.113%	<u>1.300%</u>	n/a	1.300%
Median	0.095%	0.090%	0.020%	0.016%	0.195%	0.096%	0.078%

Company	Probability of Default 2006						Models Median PD
	RAPD	Altman Z-Score	Moody's KMV	Bloomberg DRSK	Rating S&P	CDS Implied PD	
Exxon Mobil	0.001%	0.000%	n/a	0.001%	0.000%	n/a	0.001%
Texas Instruments	0.001%	0.000%	n/a	0.005%	0.090%	n/a	0.003%
Intel	0.001%	0.000%	n/a	0.002%	0.070%	n/a	0.001%
Nestlé	0.010%	0.090%	n/a	0.000%	0.000%	0.058%	0.010%
Abbott Laboratories	0.001%	0.020%	n/a	0.006%	0.020%	n/a	0.013%
Procter & Gamble	0.001%	0.090%	n/a	0.003%	0.040%	n/a	0.022%
British American Tobacco	0.001%	0.230%	n/a	0.000%	0.160%	0.112%	0.112%
Altria	0.005%	0.090%	n/a	0.005%	0.160%	0.298%	0.090%
Amphenol	0.460%	0.000%	n/a	0.146%	0.380%	n/a	0.263%
PVH	2.755%	0.090%	n/a	0.121%	<u>0.800%</u>	n/a	0.461%
TDK	0.001%	0.000%	n/a	n/a	0.040%	n/a	0.001%
Merck & Co.	0.001%	0.230%	n/a	0.014%	0.040%	n/a	0.027%
Ibm	0.090%	0.550%	n/a	0.016%	0.070%	0.071%	0.071%
Dell	0.001%	0.020%	n/a	0.001%	0.090%	n/a	0.011%
Chevron	0.420%	0.090%	n/a	0.016%	0.020%	n/a	0.055%
Xilinx	0.001%	0.000%	n/a	0.001%	0.380%	n/a	0.001%
Lockheed Martin	0.750%	0.230%	n/a	0.004%	0.160%	n/a	0.195%
Corning	0.010%	0.230%	n/a	0.012%	0.380%	n/a	0.121%
France Telecom	0.140%	27.390%	n/a	0.082%	0.120%	0.185%	0.140%
Intl Rectifier	0.001%	0.090%	n/a	0.047%	<u>0.800%</u>	n/a	0.069%
Gannett	0.030%	0.000%	n/a	0.112%	0.120%	n/a	0.071%
STM Microelectronics	0.410%	0.090%	n/a	0.000%	0.120%	0.108%	0.108%
Exelon	0.001%	9.120%	n/a	0.034%	0.160%	n/a	0.097%
Valero Energy	0.285%	0.120%	n/a	0.072%	0.380%	0.201%	0.201%
Universal Health Services	0.015%	0.040%	n/a	0.010%	0.230%	n/a	0.028%
Sealed Air	0.085%	0.230%	n/a	0.020%	0.230%	0.245%	0.230%
Oxford Industries	1.980%	0.020%	n/a	0.122%	<u>1.300%</u>	n/a	0.711%
NVR	0.005%	0.000%	n/a	0.046%	0.380%	n/a	0.026%
Boeing	4.550%	27.390%	n/a	0.046%	0.090%	0.059%	0.090%
Ryder System	0.070%	5.880%	n/a	0.141%	0.230%	n/a	0.185%
Sempra Energy	0.620%	27.390%	n/a	0.023%	0.160%	0.165%	0.165%
Safeway	2.350%	5.880%	n/a	0.077%	0.380%	0.171%	0.380%
Steel Dynamics	0.370%	0.090%	n/a	0.084%	<u>0.800%</u>	n/a	0.230%
Westlake Chemicals	0.001%	0.090%	n/a	0.088%	<u>0.800%</u>	n/a	0.089%
Toshiba	3.495%	27.390%	n/a	n/a	0.380%	n/a	3.495%
Quicksilver	3.005%	0.230%	n/a	0.457%	<u>0.800%</u>	n/a	0.629%
Shiloh Industries	0.295%	0.230%	n/a	0.326%	<u>1.300%</u>	n/a	0.310%
Pantry	1.200%	0.800%	n/a	0.260%	<u>2.600%</u>	n/a	1.000%
Vail Resort	10.830%	27.390%	n/a	0.027%	<u>1.300%</u>	n/a	6.065%
AES	7.905%	15.800%	n/a	0.219%	<u>2.600%</u>	n/a	5.253%
Median	0.078%	0.090%	n/a	0.025%	0.195%	0.165%	0.103%

Data source: Bloomberg, Moody's KMV, Standard & Poor's. – RAPD Processing: by wiredex and value.corporate [Bloomberg APPS VCORP<GO>].

Tab. 4: PD Estimates Performing Companies 2009-2011

Company	Probability of Default 2011						Models Median PD
	RAPD	Altman Z-Score	Moody's KMV	Bloomberg DRSK	Rating S&P	CDS Implied PD	
Singapore Techn. Engineering	0.001%	0.800%	0.040%	0.000%	0.000%	n/a	0.001%
Microsoft	0.001%	0.020%	0.080%	0.002%	0.000%	n/a	0.002%
Sabmiller	0.015%	0.090%	0.020%	0.000%	0.160%	n/a	0.020%
Sanofi-ADR	0.001%	0.090%	0.030%	0.001%	0.040%	0.363%	0.035%
Dr Pepper Snapple Group	0.020%	0.040%	0.200%	0.030%	0.230%	n/a	0.040%
Anheuser-Bush-Inbev	0.001%	0.230%	0.030%	0.045%	0.160%	n/a	0.045%
Louis Vuitton	0.001%	0.090%	0.010%	0.005%	0.120%	0.340%	0.050%
United Technologies	0.005%	0.090%	0.060%	0.007%	0.090%	0.237%	0.075%
Bayer AG	0.001%	0.230%	0.030%	0.015%	0.120%	0.198%	0.075%
BMC Software	0.001%	0.090%	0.080%	0.005%	0.230%	n/a	0.080%
Cisco Systems	0.001%	0.090%	0.090%	0.021%	0.070%	0.384%	0.080%
Southern Co.	0.001%	2.600%	0.070%	0.005%	0.090%	0.308%	0.080%
General Dynamics	0.005%	0.090%	0.180%	0.023%	0.090%	0.237%	0.090%
Weston (George) LTD	0.001%	0.230%	0.090%	0.082%	0.230%	n/a	0.090%
Saia Inc.	0.315%	0.000%	0.100%	0.010%	0.120%	n/a	0.100%
Owens & Minor	0.120%	0.000%	0.190%	0.024%	0.380%	n/a	0.120%
Duke Energy	0.135%	2.600%	0.090%	0.008%	0.120%	0.144%	0.128%
Raytheon	0.040%	0.230%	0.170%	0.049%	0.120%	0.256%	0.145%
Spectra Energy Corp.	1.205%	5.880%	0.120%	0.072%	0.160%	n/a	0.160%
Waste Management	1.350%	1.300%	0.080%	0.008%	0.230%	0.530%	0.380%
Mylan	0.415%	0.800%	0.120%	0.042%	<u>0.800%</u>	n/a	0.415%
Molson Coors Brewing	1.175%	0.800%	0.080%	0.126%	0.380%	0.479%	0.430%
Fujitsu Limited	1.120%	0.800%	0.620%	n/a	0.120%	n/a	0.620%
European Aeronautic Defence	1.255%	27.390%	0.700%	0.001%	0.120%	0.634%	0.667%
Con-way	21.775%	0.550%	0.840%	0.273%	0.380%	1.324%	0.695%
Windstream	0.060%	5.880%	0.370%	0.046%	<u>1.300%</u>	1.098%	0.734%
MeadWestvaco	1.490%	27.390%	0.560%	0.093%	0.230%	1.002%	0.781%
Ingles Markets	3.485%	0.800%	1.960%	0.227%	<u>1.300%</u>	n/a	1.300%
Micron Technology Inc.	0.001%	5.880%	1.390%	0.215%	<u>5.880%</u>	n/a	1.390%
Advanced Micro Devices	2.500%	27.390%	0.990%	0.622%	<u>2.600%</u>	1.423%	1.961%
Consolidated Communications	26.610%	2.600%	n/a	0.091%	<u>2.600%</u>	n/a	2.600%
Alaska Comm. Systems Group	25.735%	27.390%	1.310%	0.154%	<u>2.600%</u>	n/a	2.600%
Valhi	14.280%	5.880%	2.420%	0.661%	<u>5.880%</u>	n/a	5.880%
Median	0.060%	0.550%	0.110%	0.027%	0.160%	0.374%	0.128%

Company	Probability of Default 2010						Models Median PD
	RAPD	Altman Z-Score	Moody's KMV	Bloomberg DRSK	Rating S&P	CDS Implied PD	
Singapore Techn. Engineering	0.001%	0.230%	0.040%	0.000%	0.000%	n/a	0.001%
Microsoft	0.001%	0.000%	0.070%	0.069%	0.000%	n/a	0.001%
Sabmiller	0.195%	0.230%	0.040%	0.040%	0.160%	n/a	0.160%
Sanofi-ADR	0.001%	0.090%	0.020%	0.004%	0.040%	0.472%	0.030%
Dr Pepper Snapple Group	0.025%	0.040%	0.240%	0.092%	0.380%	n/a	0.092%
Anheuser-Bush-Inbev	0.001%	0.230%	0.030%	0.059%	0.160%	n/a	0.059%
Louis Vuitton	0.001%	0.090%	0.020%	0.020%	0.120%	0.490%	0.055%
United Technologies	0.030%	0.230%	0.070%	0.034%	0.090%	0.311%	0.080%
Bayer AG	0.001%	0.230%	0.040%	0.069%	0.120%	0.429%	0.095%
BMC Software	0.005%	0.090%	0.100%	0.073%	0.230%	n/a	0.090%
Cisco Systems	0.001%	0.020%	0.050%	0.092%	0.070%	0.275%	0.060%
Southern Co.	0.001%	2.600%	0.100%	0.051%	0.090%	0.472%	0.095%
General Dynamics	0.020%	0.090%	0.140%	0.049%	0.090%	0.372%	0.090%
Weston (George) LTD	0.001%	0.090%	0.090%	0.046%	0.230%	n/a	0.090%
Saia Inc.	0.300%	0.000%	0.120%	0.061%	0.120%	n/a	0.120%
Owens & Minor	0.355%	0.000%	0.230%	0.077%	0.380%	n/a	0.230%
Duke Energy	0.255%	2.600%	0.120%	0.040%	0.120%	0.347%	0.188%
Raytheon	0.005%	0.090%	0.130%	0.056%	0.120%	0.439%	0.105%
Spectra Energy Corp.	0.770%	5.880%	0.170%	0.081%	0.160%	n/a	0.170%
Waste Management	0.435%	0.800%	0.120%	0.060%	0.230%	0.501%	0.333%
Mylan	1.250%	5.880%	0.320%	0.148%	<u>0.800%</u>	n/a	0.800%
Molson Coors Brewing	1.440%	0.800%	0.100%	0.523%	0.380%	0.446%	0.485%
Fujitsu Limited	2.995%	5.880%	2.730%	n/a	0.120%	n/a	2.730%
European Aeronautic Defence	5.585%	27.390%	0.920%	0.061%	0.160%	0.907%	0.913%
Con-way	17.075%	2.600%	1.520%	0.590%	0.380%	1.966%	1.743%
Windstream	0.001%	0.550%	0.380%	0.185%	<u>0.800%</u>	2.372%	0.465%
MeadWestvaco	28.135%	27.390%	0.960%	0.180%	0.230%	0.683%	0.821%
Ingles Markets	2.495%	0.550%	4.110%	0.801%	<u>1.300%</u>	n/a	1.300%
Micron Technology Inc.	99.715%	27.390%	0.890%	0.269%	<u>5.880%</u>	n/a	5.880%
Advanced Micro Devices	97.890%	100.000%	1.680%	0.307%	<u>9.120%</u>	3.315%	6.217%
Consolidated Communications	19.385%	5.880%	n/a	0.269%	<u>2.600%</u>	n/a	4.240%
Alaska Comm. Systems Group	42.025%	27.390%	2.950%	0.379%	<u>2.600%</u>	n/a	2.950%
Valhi	100%	27.390%	4.870%	1.582%	<u>27.390%</u>	n/a	27.390%
Median	0.255%	0.550%	0.125%	0.071%	0.160%	0.472%	0.170%

Company	Probability of Default 2009						Models Median PD
	RAPD	Altman Z-Score	Moody's KMV	Bloomberg DRSK	Rating S&P	CDS Implied PD	
Singapore Techn. Engineering	0.065%	0.800%	0.070%	0.000%	0.000%	n/a	0.065%
Microsoft	0.001%	0.020%	0.030%	0.002%	0.000%	n/a	0.002%
Sabmiller	0.020%	0.230%	0.020%	0.003%	0.160%	n/a	0.020%
Sanofi-ADR	0.001%	0.020%	0.020%	0.024%	0.040%	1.609%	0.022%
Dr Pepper Snapple Group	0.090%	0.230%	0.370%	0.477%	0.380%	n/a	0.370%
Anheuser-Bush-Inbev	9.010%	27.390%	0.040%	0.635%	0.160%	n/a	0.635%
Louis Vuitton	0.001%	0.230%	0.050%	0.220%	0.120%	2.786%	0.170%
United Technologies	0.070%	0.230%	0.090%	0.254%	0.090%	1.240%	0.160%
Bayer AG	0.345%	0.800%	0.040%	0.564%	0.120%	1.967%	0.455%
BMC Software	0.001%	0.230%	0.040%	0.012%	0.230%	n/a	0.040%
Cisco Systems	0.001%	0.090%	0.020%	0.018%	0.070%	0.390%	0.045%
Southern Co.	0.001%	2.600%	0.060%	0.135%	0.090%	1.775%	0.122%
General Dynamics	0.050%	0.090%	0.160%	0.169%	0.090%	1.211%	0.125%
Weston (George) LTD	0.220%	0.230%	0.070%	0.167%	0.230%	n/a	0.220%
Saia Inc.	0.230%	0.000%	0.130%	0.029%	0.120%	n/a	0.120%
Owens & Minor	1.440%	0.000%	0.130%	0.140%	0.380%	n/a	0.140%
Duke Energy	0.005%	5.880%	0.110%	0.219%	0.120%	0.923%	0.170%
Raytheon	0.045%	0.230%	0.070%	0.189%	0.120%	0.807%	0.154%
Spectra Energy Corp.	0.685%	9.120%	0.270%	0.887%	0.160%	n/a	0.685%
Waste Management	0.270%	2.600%	0.070%	0.138%	0.230%	1.381%	0.250%
Mylan	1.150%	27.390%	0.370%	0.558%	<u>1.300%</u>	n/a	1.150%
Molson Coors Brewing	2.030%	0.230%	0.090%	0.181%	0.230%	1.193%	0.230%
Fujitsu Limited	0.020%	5.880%	0.800%	n/a	0.120%	n/a	0.160%
European Aeronautic Defence	0.250%	27.390%	0.780%	0.043%	0.160%	5.398%	0.515%
Con-way	0.740%	1.300%	1.580%	1.398%	0.380%	6.015%	1.349%
Windstream	0.001%	0.800%	0.380%	0.459%	<u>0.550%</u>	2.118%	0.504%
MeadWestvaco	21.63%	27.390%	1.720%	1.610%	0.230%	2.790%	2.255%
Ingles Markets	1.095%	0.550%	1.360%	0.938%	<u>1.300%</u>	n/a	1.095%
Micron Technology Inc.	98.31%	100.000%	1.330%	1.010%	<u>1.300%</u>	n/a	1.330%
Advanced Micro Devices	99.980%	100.000%	15.610%	7.724%	<u>5.880%</u>	46.707%	31.158%
Consolidated Communications	9.645%	9.120%	n/a	0.955%	<u>2.600%</u>	n/a	5.860%
Alaska Comm. Systems Group	16.270%	27.390%	2.130%	0.699%	<u>2.600%</u>	n/a	2.600%
Valhi	87.690%	27.390%	2.830%	2.061%	<u>5.880%</u>	n/a	5.880%
Median	0.230%	0.800%	0.120%	0.220%	0.160%	1.692%	0.230%

Data source: Bloomberg, Moody's KMV, Standard & Poor's. – RAPD Processing: by wiredex and value.corporate [Bloomberg APPS VCORP<GO>].

Appendix A: RAPD back-testing comparative analysis assumptions

RAPD Approach implementation hypotheses

The assumptions described here below for the implementation of the RAPD approach were developed exclusively for the purpose of ex-post comparison of the performance of various credit risk models, and thus respond to the specific needs and problems of this type of test. Therefore, these hypotheses do not by any means represent the only or the best implementation paradigm for the RAPD model. For other analysis needs, users can easily adopt more evolved implementation paradigms capable of more efficaciously measuring credit risk; professional analysts who are interested in applying the RAPD Approach can and should make use of all the most updated information available and the assumptions that best fit with the case in point. However, the use of ad hoc hypotheses necessary for the application of more evolved paradigms may seem “suspicious” within the context of an ex-post comparative back-testing analysis. Hence the need to adopt standard hypotheses applied equally to all of the companies analyzed, as simplified and neutral as possible, albeit in full awareness of their limitations.

- *Number of simulation trials*
20,000 trials were generated for each simulation process.
 - *Forecast period*
The explicit forecast period always considers three years. It is presumed that PDs are estimated at the end of the month following the fiscal year-end month, which in most cases is December. Therefore, for example, supposing a December closing, PD for the year 2008 is considered as if it were estimated on January 31st, 2008, with balance sheet data from December 31st, 2007 available; the PD estimate thus obtained is maintained constant for the entire year.
 - *Probability distributions*
For all of the stochastic variables we used Weibull distribution functions, which are modeled depending on the type of variable. We decided to employ one type of distribution function for all the stochastic variables in order to simplify the analysis process. The choice of the Weibull function and its parameter setting for each stochastic variable arises from empirical cross-section analysis we performed over the industry sectors to which the companies analyzed belonged, aimed at finding the best-fitting distribution function.
 - *Forecast variables*
For stochastic forecast variables, the following hypotheses apply:
 - Revenues: forecast in terms of rate of variation. To model the variable’s probability distribution, a Weibull (shape parameter: 2) with positive asymmetry was used. The distribution function was set by establishing the average and the minimum values of the distribution. The average distribution value is held constant for all three explicit forecast years, and is set equal to the last five years average of the nominal GDP growth rate in the company’s reference country at the time of estimation. The minimum value is determined on the basis of data relative to a sample of comparable companies from the same GICS³³ industry sector (at sub-industry level) in the three years prior to that in which the PD estimate is presumed to occur. The minimum distribution value is determined by subtracting from the average value a difference based on the company’s GICS industry sector data recorded in the three years prior to that in which the PD estimate is presumed to occur:
$$[company\ minimum\ distribution] = [company\ average] - [sector\ difference].$$

The sector difference has been determined as the maximum difference in the three years between the sector average and minimum revenue rate of variation. Sector outliers have been cleared from the sample by eliminating the lowest revenue rate of variation below a given percentile of the records registered.
 - Operating costs (EBITDA margin): considered in the aggregate and forecast as a percentage of revenues. The distribution function assumed is a Weibull (shape parameter: 3.5) that basically replicates a symmetric distribution function. The distribution function was set by establishing the average and minimum values of the function. The average is set equal to the value of the operating cost percentage over revenues arising from the last available financial statement; the minimum follows the same rules described for revenues. These modeling assumptions imply that the more the firm’s current performance departs from the best performing companies within the industry, the higher the probabilities that the firm will work with high operating costs.
 - Investments in tangible and intangible assets (Capex) are calculated implicitly on the basis of the ratio between net fixed asset value (after any goodwill items) and value of revenues³⁴. The distribution function assumed is a Weibull function (shape parameter: 3.5), set by establishing the average and minimum values of the function. The average is set equal to the value of the ratio arising from the last available financial statement; the minimum follows the same rules described for revenues operating.
 - Net working capital funds and other: includes inventories, accounts receivable and payable, other current and non-current assets/liabilities. They are forecasted in terms of ratio respect to revenues; the distribution function assumed is a Weibull (shape parameter: 3). The distribution function was set by establishing the average and minimum values of the function. The average is set equal to the last available financial statement; the minimum follows the same rules described for revenues and operating cost.
- All other non-stochastic variable assumptions are made on the basis of the following rules:
- Financial debts and liquidity are calculated considering a financial equilibrium constraint as hypothesized by the RAPD model (see paragraph 2.1).
 - Interest rates on debt are determined as a function of risk-free rates increased by a credit spread, determined on the basis of the company’s rating at the time of analysis. The yield curve of the risk-free rates and the credit spread with which cost of debt is calculated refer to the country and the time at which valuation is presumed to have occurred.

³³ GICS (Global Industry Classification Standard), see: <http://www.msci.com/products/indices/sector/gics/> and <http://www.standardandpoors.com/products-services/GICS/en/us>.

³⁴ We used a procedure quite similar to that illustrated in footnote 7.

- We hypothesize that dividends are not distributed, and all net cash flow generated that does not go to repay debt remains in the company and generates, in each period, a yield equal to risk-free rates implicit in the yield curve at the time when PD estimation is presumed to have occurred.
- Taxes are determined on the basis of the company's effective tax rate of the last available financial statement; the effective tax rate is normalized by constraining it within a range given by a maximum tax rate (150% of the company's country tax rate) and a minimum tax rate (70% of the company's country tax rate).
- *Correlation matrix*
The following standard hypotheses of correlation were considered:
 - $\text{Autocorr}[\text{Sales}_{(t)}, \text{Sales}_{(t-1)}] = 0.2$
 - $\text{Autocorr}[\text{Operating Cost}/\text{Sales}_{(t)}, \text{Operating Cost}/\text{Sales}_{(t-1)}] = 0.3$
 - $\text{Autocorr}[\text{Net Fixed Asset}/\text{Sales}_{(t)}, \text{Net Fixed Asset}/\text{Sales}_{(t-1)}] = 0.5$
 - $\text{Autocorr}[\text{Net Working Capital Funds \& Other}_{(t)}, \text{Net Working Capital Funds \& Other}_{(t-1)}] = 0.4$
 - $\text{Corr}[\text{Sales}_{(t)}, \text{Operating Cost}/\text{Sales}_{(t)}] = -0.4$
 - $\text{Corr}[\text{Net Fixed Asset}/\text{Sales}_{(t)}, \text{Operating Cost}/\text{Sales}_{(t)}] = -0.2$
 - $\text{Corr}[\text{Sales}_{(t)}, \text{Net Fixed Asset}/\text{Sales}_{(t)}] = 0.2$
 - $\text{Corr}[\text{Sales}_{(t)}, \text{Net Working Capital Funds \& Other}_{(t)}] = -0.3$
- *Enterprise Value*
In each scenario of the simulation, EV is calculated through the Capital Cash Flow model (see Ruback, 2002), as the sum of the present value of operating cash flows (defined in [3]) within the explicit forecast period, plus a terminal value obtained as perpetuity of Nopat (Net Operating Profit After Tax) + Interest Tax Shield of the last explicit forecast period. As cost of capital we applied a pre-tax wacc, calculated considering as the risk-free rate the 10-year Government Bond at the time of the valuation, a market risk premium of 5% and an unlevered beta obtained as the median value of the unlevered beta coefficients within the company's GICS industry sector.

Data sources

- *Bloomberg*
Historical balance sheet data, interest rates, credit spread, market prices, S&P ratings, CDS spread and PD estimates denoted as Bloomberg DRSK.
- *Moody's KMV*
PD estimates denoted as Moody's KMV EDFTM were obtained based on the Moody's KMV model. Keep in mind that the PD scale shown for the Moody's KMV model has a cap value of 20% for estimates until 2007, after which the cap is 35%.
- *Altman Z-Score*
We used Z-Score for manufacturing companies based on the following Altman model:

$$Z = 1.2 \cdot \frac{\text{Working Capital}}{\text{Total Assets}} + 1.4 \cdot \frac{\text{Retained Earnings}}{\text{Total Assets}} + 3.3 \cdot \frac{\text{EBIT}}{\text{Total Assets}} + 0.6 \cdot \frac{\text{Market Value of Equity}}{\text{Book Value Total Liabilities}} + 1.0 \cdot \frac{\text{Sales}}{\text{Total Assets}}$$

For non-manufacturing companies we use the Z''-Score based on the following Altman model:

$$Z = 3.25 + 6.56 \cdot \frac{\text{Current Asset} - \text{Current Liabilities}}{\text{Total Assets}} + 3.26 \cdot \frac{\text{Retained Earnings}}{\text{Total Assets}} + 6.72 \cdot \frac{\text{EBIT}}{\text{Total Assets}} + 1.05 \cdot \frac{\text{Book Value of Equity}}{\text{Total Liabilities}}$$

The Z-Score is transformed into PD through a dual conversion process: (1) with a conversion table developed by Altman, the Z-Score is translated into an equivalent rating class, see Altman and Hotchkiss (2006), p. 247 for Z-Score and p. 248 for Z''-Score; (2) the rating is converted into PD using historical frequencies of default produced by rating agencies (see the following point).
- *Conversions [Rating $\rightleftharpoons$ PD]*
Conversion from PD to Rating and vice-versa is obtained from "Cumulative Average Default Rates" tables, based on historical frequencies of default recorded in various rating classes between 1981 and 2010; see Standard and Poors (2011). To reconstruct the master scale, the historical default tables of rated companies, provided by rating agencies, were utilized. Although this is the only practicable method, as well as the one universally adopted, it is clear that this process presents approximations and limitations that may alter the translation of ratings and scores into PDs. To better understand the type of approximation, consider that the historical default tables provided by rating agencies merely record the frequency of default which after 1, 2, ..., n years occurs in each rating class among companies that had said rating class in the initial period of reference. But often, especially for the best rating classes, the sample of companies of reference can be quite small, and thus provided a statistically unreliable measurement. Furthermore, again due to the limited number of cases of default, it is impossible to obtain a frequency for every rating class and sub-class. We also find gaps or leaps in empirical measurements of default between one class and another (this occurs in particular for the worst rating classes, far below investment grade). Consequently, the chart of historical frequencies of default presents some inconsistent results – for example, for some rating classes (such as A), it provides frequencies of default that are higher than those for worse rating classes (such as A-). To obviate these drawback, we have normalized the effective historical frequencies by means of functions that reconstruct (approximating) the curve of historical PDs, so as to eliminate inconsistent results. The following table reports the master scale used to convert ratings in-

to 1-year PDs.

AAA	AA+	AA	AA-	A+	A	A-	BBB+	BBB	BBB-	BB+
0.000%	0.010% ^(*)	0.020%	0.040%	0.070%	0.090%	0.120% ^(*)	0.160%	0.230%	0.380%	0.550%
BB	BB-	B+	B	B-	CCC+	CCC	CCC-	CC	C	D
0.800%	1.300%	2.600%	5.880%	9.120%	15.805% ^(**)	27.390%	37.861% ^(**)	52.335% ^(**)	72.343% ^(**)	100.000%

^(*) PDs adjusted by exponential interpolation because inconsistent. ^(**) PDs created by exponential interpolation due to lack of data.

- *Conversions [CDS Spread → PD]*

CDS Spread is transformed into PD through the following formula:

$$PD = 1 - e^{-\left(CDS/LGD\right)}$$

Where LGD represents a loss given default with a Recovery Rate of 40%.

Appendix B: RAPD recovery rate estimates for the defaulted companies sample

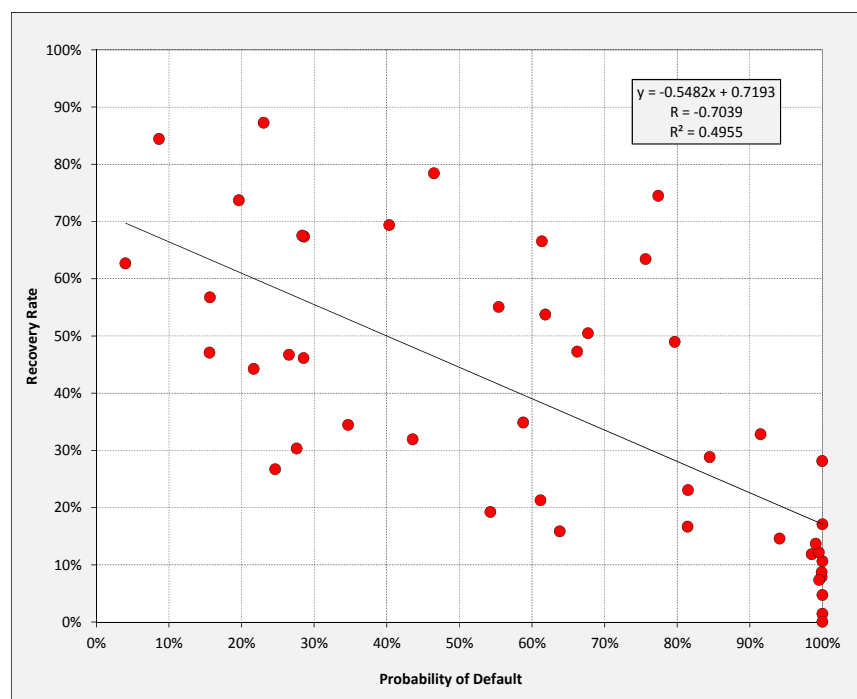
Tab. B1: Defaulted Companies Unsecured Recovery Rates

Company	Date of Default (m-y)	Recovery Rate	Company	Date of Default (m-y)	Recovery Rate
Kevco	Feb-01	15.84%	Dana	Mar-06	23.06%
Polaroid	Jul-01	46.71%	Dura	Oct-06	67.35%
Newcor	Sep-01	55.09%	Seacontainer	Oct-06	28.15%
Swissair	Oct-01	34.47%	Tousa	Jan-08	47.08%
Federal Mogul	Oct-01	84.41%	Tronox	Mar-08	28.83%
Enron	Dec-01	26.72%	Pilgrim's Pride	Nov-08	30.32%
IT Group	Jan-02	44.24%	VeraSun Energy	Nov-08	63.44%
Covanta Energy	Mar-02	53.74%	Tribune	Dec-08	46.14%
IFCO Systems	Mar-02	0.04%	Smurfit-Stone Cont.	Feb-09	56.74%
Exide Technologies	Apr-02	50.48%	Chemtura	Mar-09	31.92%
Worldcom	Jul-02	66.54%	AbitibiBowater	Apr-09	7.88%
Energis	Aug-02	13.72%	Spectrum Brands	Apr-09	69.38%
Amerco	Oct-02	74.48%	Six Flags Inc.	Apr-09	78.43%
Gen Tek	Oct-02	32.83%	Visteon	May-09	16.68%
Key3Media	Dec-02	48.95%	Ion Media Networks	May-09	17.07%
United Airlines	Dec-02	1.45%	Citadel Broadcasting	Dec-09	67.52%
Qwest Commun.	Dec-02	12.16%	Newpage	Jul-11	8.73%
Healthsouth	Apr-03	73.71%	AMR	Nov-11	34.87%
Air Canada	Apr-03	14.58%			
Mississippi Chemical	May-03	7.37%			
Mirant	Jul-03	87.27%			
Parmalat	Dec-03	62.69%			
Solutia	Dec-03	47.26%			
Winn-Dixie	Feb-05	19.22%			
Northwest	Sep-05	11.87%			
Delphi	Oct-05	21.28%			
Calpine	Dec-05	10.64%			

Statistics Full Sample	
Minimum	0.04%
Average	37.96%
Median	33.65%
Maximum	87.27%
Standard Deviation	25.01%

Data source: Bloomberg. – RAPD Processing: by wiredex and value.corporate [Bloomberg APPS VCORP<GO>].

Fig. B1: Unsecured Recovery Rate and Default Probability



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